

**Dubai Metro, UAE
Green Line – Al-Qusais Depot**

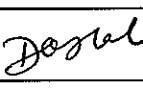
**Vehicle Repair
(Building 18)**

**Structural Design Calculations
(DCP3 Submission)**

Aug 2007

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
Document No.: DM001 -D-GAP-WMD-DR-DCC-544999	Controlled Copy No.:
Document Title: Green Line – Al-Qusais Depot Vehicle Repair (Building 18) Structural Design Calculations (DCP3 Submission)	
Covering Letter / Transmittal Ref. No.:	Date of Issue: 12 Aug 2007

Revision, Review and Approval/Authorization Records

				
A1	DCP3 Submission	D. Shah Aug 07	P. Scott Aug 07	J. Hill Aug 07
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

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3	Atkins (Paul Shepherd-Smith)
4	Atkins, (Dipak Shah)

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1 Introduction

The Dubai Metro Project is under construction by the Dubai Rapid Link Consortium (DURL). The Civil Works are contracted to the Japan Turkey Metro Joint Venture (JTMJV), and the Rail Systems Works are contracted to Mitsubishi Heavy Industries (MHI). The rolling stock will be provided by Kinki Sharyo.

The vehicle repair workshop will be housed in a building 19m by 20m and 6m high and undertake all repair and maintenance for the operational and maintenance vehicle fleet.

This submission for Vehicle Repair (Building 18) in Al-Qusais Depot is made by the Designers for the JTMJV, and as such concentrates on the scope of the Civil Works Contract, which includes the Architecture, Structure and M&E Design.

2 Building Form and Structural System

The vehicle repair workshop will be housed in a building 19m long, 20m wide and 6m high. Proposed building is identical as that of Red Line Rashidiya Main Depot, Building 23, Vehicle Repair Workshop.

The building has a structural grid of 8 m x 10 m which is stable and most suitable to house the facilities. Two bays of 10 m span portal frame with I-Beam @ 5 m c/c in transverse direction is proposed to support the roof with cold form Z shape steel section purlin spaced at 1.5 m c/c resting above the beam. Diagonal bracing system, circular rods are used for bracing of roof members.

A 300 mm thick Ground level slab monolithic with Reinforced Concrete Beam is proposed to house the facilities. The RC Pedestal raised from RC pad footing will support structural steel columns with holding down bolts fixed in position at formation level.

3 Peer Review of Red Line building and comparison

Essentially this building is identical as that of Red Line Rashidiya Main Depot, Building 23 Vehicle Repair Workshop. So the form and structural system has been replicated from the Rashidiya building.

The structural report DM001-E-MAP-CDP-DR-DCC-371951-A4 (Ref. Appendix 'A') for the Rashidiya building 23-Vehicle Repair Workshop has been peer reviewed for loadings, analysis, strength & deflection checks for all the members and foundation design. The drawings have also been peer reviewed for the design intent translation on to the details, which

have then been replicated onto the Green Line Depot building-18-Vehicle Repair as listed in sect.4.

The basis of this report is to utilise Red Line Main Depot Vehicle Repair Workshop (Bldg-23) to Green Line Al-Qusais Depot Vehicle Repair (Bldg-18) with sufficient review to allow submission of Green Line DCP3 Phase.

4 Drawings

To complement the purpose of this submission, the following Structural drawings are submitted along with this report.

	<u>Drawing Title</u>	<u>Drawing No.</u>
Structural Drawings		
1	Al-Qusais-Depot,Building-18–Vehicle Repair -Structural Drawing List	DM001-D-GAP-WMD-DD-DCC-551674
2	Al-Qusais-Depot - General Notes And Typical Details - Sheet 1 Of 3	DM001-D-GDP-CDP-DD-DCC-551990
3	Al-Qusais-Depot - General Notes And Typical Details - Sheet 2 Of 3	DM001-D-GDP-CDP-DD-DCC-551991
4	Al-Qusais-Depot - General Notes And Typical Details - Sheet 3 Of 3	DM001-D-GDP-CDP-DD-DCC-551992
5	Al-Qusais-Depot,Building-18–Vehicle Repair - Foundation Layout Plan	DM001-D-GAP-WMD-DD-DCC-551650
6	Al-Qusais-Depot,Building-18–Vehicle Repair - Ground Slab Plan	DM001-D-GAP-WMD-DD-DCC-551651
7	Al-Qusais-Depot,Building-18–Vehicle Repair - Roof Layout Plan	DM001-D-GAP-WMD-DD-DCC-551652
8	Al-Qusais-Depot,Building-18–Vehicle Repair - Section A-A & B-B	DM001-D-GAP-WMD-DD-DCC-551660
9	Al-Qusais-Depot,Building-18–Vehicle Repair - Section C-C	DM001-D-GAP-WMD-DD-DCC-551661
10	Al-Qusais-Depot,Building-18–Vehicle Repair - Footing Details	DM001-D-GAP-WMD-DD-DCC-551662

	<u>Drawing Title</u>	<u>Drawing No.</u>
11	Al-Qusais-Depot, Building- 18 – Vehicle Repair – Beams/Slab/Pit Reinforcement Details	DM001-D-GAP-WMD-DD-DCC-551663
12	Al-Qusais-Depot,Building-18–Vehicle Repair–Roof Setting Out – Plan & Sections	DM001-D-GAP-WMD-DD-DCC-551664

Appendix A
Red Line Rashidiya Depot
Vehicle Repair Building - 18
Design Calculations (Enhanced Design)

**Dubai Metro, UAE
Rashidiya Depot**

**Structural Design Calculation
(Enhanced Design)
Building 23 – Vehicle Maintenance Shop**

May 2007

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
Document No.: DM001-E-MAP-CDP-DR-DCC-371951	Controlled Copy No.:
Document Title: Main Depot- Rashidiya Structural Design Calculation (Enhanced Design) Building 23 – Vehicle Maintenance Shop	
Covering Letter / Transmittal Ref. No.: To be advised	Date of Issue: 02 May 2007

Revision, Review and Approval/Authorization Records

A4	Issued for Approval (ENHANCED DESIGN)	CK Lam May 07	P. Shepherd-Smith May 07	B. Maney May 07
A3	Issued for Approval	CK Lam Nov 06	P. Shepherd-Smith Nov 06	B. Maney Nov 06
A2	Issued for Approval	CK Lam Nov 06	P. Shepherd-Smith Nov 06	B. Maney Nov 06
A1	Internal Review	CK Lam Oct 06	P. Shepherd-Smith Oct 06	B. Maney Oct 06
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1 TYPICAL PORTAL FRAME TRANSVERSE ANALYSIS

Document No.: **DM001-E-MAP-CDP-DR-DCC-371951**

Rev A4

Date: 02 May 07

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Job No

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Job Title

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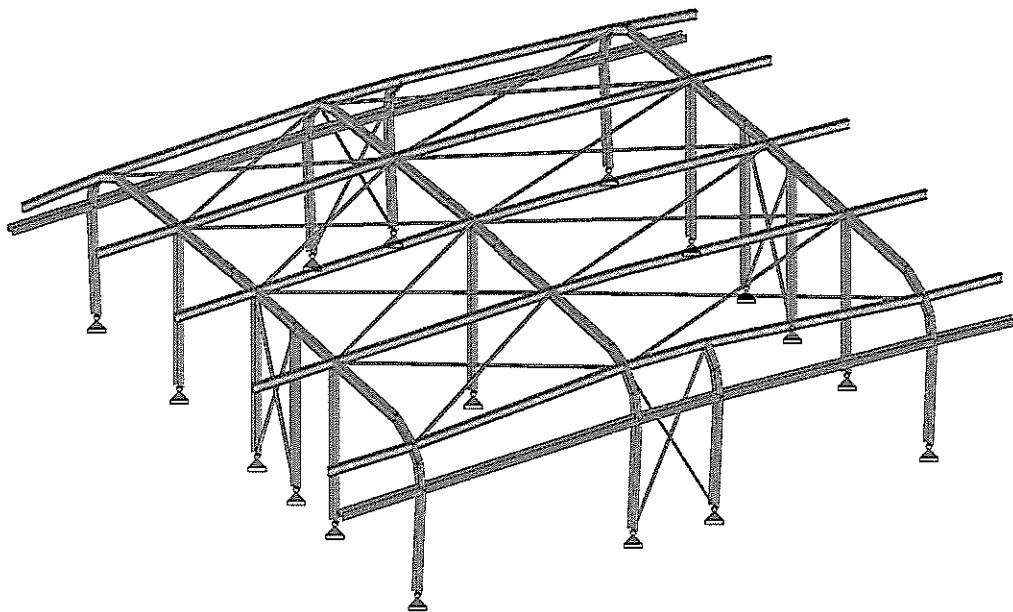
Date April 2007

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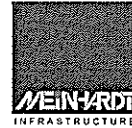
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Date/Time 28-Apr-2007 14:50



Load 1

VEHICLE MAINTENANCE SHOP



Site:	Dubai Metro Project	Date:	28-Apr-07	Job No.:	68010.01
	Vehicle Maintenance Shop				
	Loadings	Designed by:	Rudy	Sheet No.:	

Roof Loadings :**DL** W_{DL} :

Sheeting	0.1 kPa		
Purlins	0.1 kPa		
Insulation	0.3 kPa		
M&E Finishes	0.3 kPa		
	<u>0.8 kPa</u>	x 1 =	0.8 kN/m

LL

0.6 kPa	x 1 =	0.6 kN/m
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P_{LL} : A/C equipment S/W
 = 5 kN

Notional horizontal load, Ph

(Load Case of 1.4DL + 1.6LL)

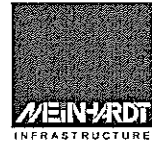
Tributary load area for the roof
 = $(19+2*3)*(20+2*1.5)$
 = 575 m²

Total Dead Load
 460 kN

Total Live Load
 350 kN
 (Including A/C equipment)

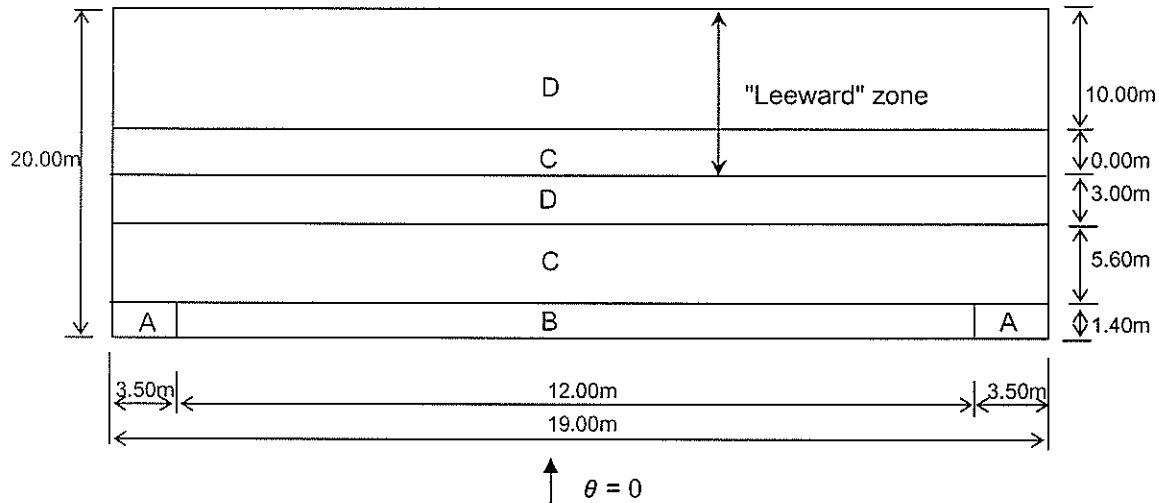
Tributary load area for the wall
 = $7*2*(19+2*3+20+2*1.5)$
 = 672 m²

Total Dead Load
 336 kN
 (excluding the M&E Finishes)



Site:	Dubai Metro Project	Date:	28-Apr-07	Job No.:	68010.01
	Vehicle Maintenance Shop				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to eave)

PLAN

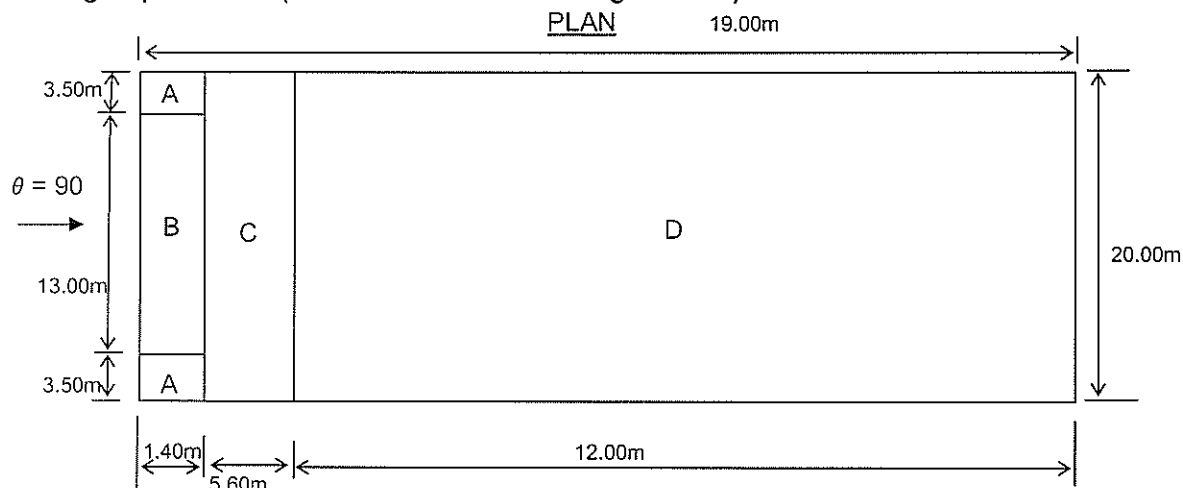
Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 19 \text{ m} \\
 H &= 7 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 7 \\
 &= 14 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m ²)	Cpe	Area	Cpe1	Cpe2
A	9.8	-2			
B	16.8	-1.4			
C	106.4	-0.7	0.0	-0.7	-0.7
D	57.0	-0.2	190.0	-0.2	0.2
Total	190.0		190.0		
Average		-0.68		-0.20	0.20



Site:	Dubai Metro Project	Date:	28-Apr-07	Job No.:	68010.01
	Vehicle Maintenance Shop				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 20 \text{ m} \\
 H &= 7 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 7 \\
 &= 14 \text{ m}
 \end{aligned}$$

For maximum uplift, assume zone C coefficient =>

$$C_{pe} = -0.7$$

For maximum downward force, assume zone D coefficient =>

$$C_{pe} = 0.2$$

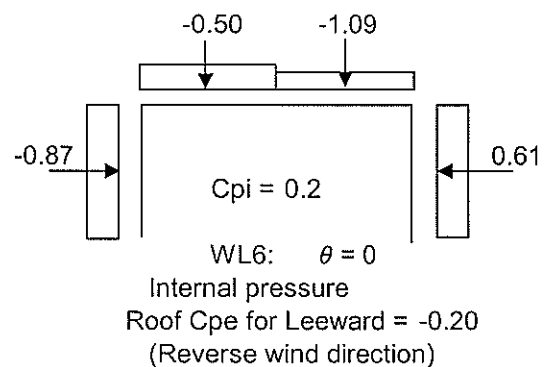
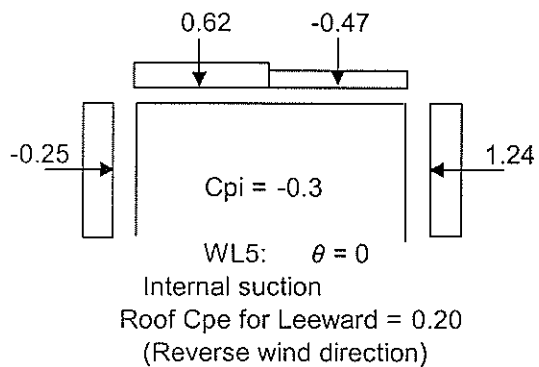
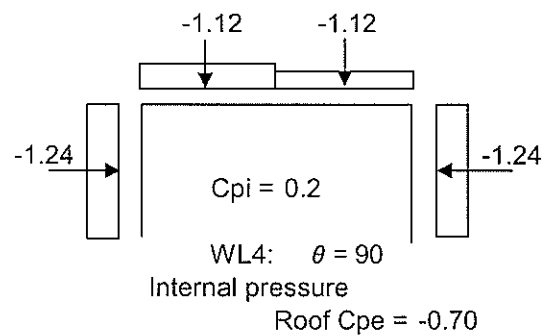
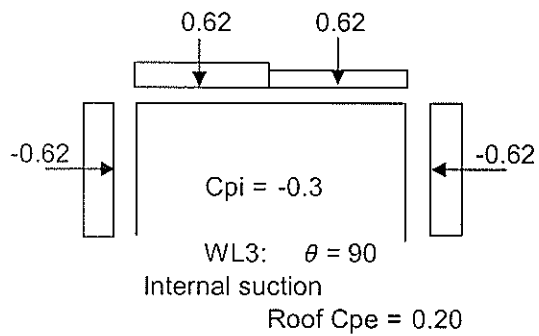
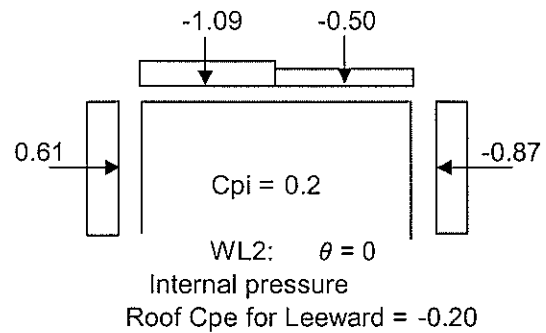
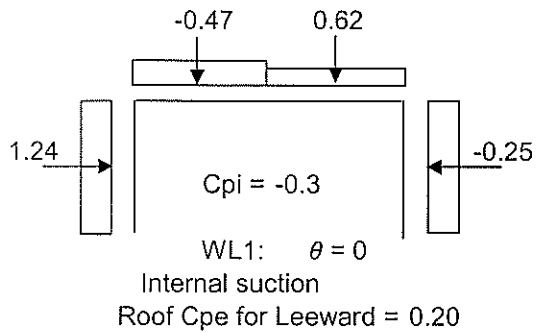
Wind Load Computation

$$\begin{aligned}
 D &= 20.00 \text{ m} \\
 H &= 7 \text{ m} \\
 D/H &= 2.86
 \end{aligned}$$

Elements	Effective Wind Speed V_e m/s	Dynamic Wind Pressure q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-0.68	-0.20 0.20	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.70	-0.5 -0.5	-0.8 (Zone B)	-0.8 (Zone B)

Site:	Dubai Metro Project Vehicle Maintenance Shop Wind Load Computation	Date:	28-Apr-07	Job No.:	68010.01
		Designed by:	Rudy	Sheet No.:	

Wind Load = 1 x q x (Cpe - Cpi) kN/m (For unit width)



Site: Dubai Metro Project

Date: 30-Apr-07

Job No.: 68010.01

Vehicle Maintenance Shop

Earthquake Loading

Designed by: Rudy

Sheet No.:

Estimation of Earthquake Loading

Total Dead Load on the Building Column W = Given below (including part of Live Load)
 Height of the Column h_n = Given below

Design Information:

- 1) Seismic Zone = 2A (from Design Brief)
 2) Occupancy Category = 1 (Table 16-K, Conservatively adopt 1)
 3) Lateral Force procedure = **Static Procedure** (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

- 4) **Base Shear** $V_{base-S} = (C_v \times I \times W) / (R \times T)$ (Section 1630.2.1)
 Seismic Coefficient $C_v = 0.32$ (Table 16-R assuming soil profile type S_D)
 Importance Factor $I = 1.00$ (Table 16-K)
 Numeric. Coef: Strength & Ductility $R = 4.50$ (Table 16-N for OMRT steel only)
 Elastic Fundamental Period of Vibration $T = C_t (h_n)^{3/4}$ Sec.
 Numerical Coefficient $C_t = 0.0853$ (Steel frame t.b.c)

- 5) **Limiting Values of Base shear** $V_{base-S-Max} = (2.5 \times C_a \times I \times W) / R = 118.68$
 $V_{base-S-Min} = 0.11 \times C_a \times I \times W = 23.50$

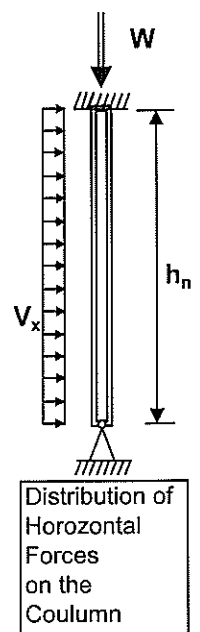
Design Earthquake Loads

- (Section 1630.6)
 $E_x = \rho \times \text{Min. } (E_h \text{ or } V_{base-S-Max}) + E_v$
 $E_{x-max} = \Omega \times \text{Min. } (E_h \text{ or } V_{base-S-Max})$
 $V_{base-S} = E_h =$ Earthquake load in the element due to the base shear, V , as set out above (for non-structural components refer to section 1632)
 Vert. component of EQ ground motion $E_v = 0.5 C_a I D$ (Table 16-K)
 Seismic Coefficient $C_a = 0.22$ (Table 16-Q, for soil type S_D)
 Seismic Force Amplification Factor $\Omega = 2.80$ (Table 16-N)
 Dead Load on the element $W = D =$ Given below
 $\rho = 2 - 6.1 / (r_{max} \sqrt{A_B}) = 1.25$ (conservative) ($1.0 \leq \rho \leq 1.5$)
 $r_{max} =$ The maximum element-storey shear ratio
 $A_B =$ Ground floor area of the structure

Distribution of Horizontal Forces:Distribution of Horizontal Force on the Column $V_x = E_x / h_n$

Col	Gr. Load, W , kN	Ht h_n , m	Time, T sec.	V_{base-S} kN	V_{base-S} kN	E_v kN	E_x kN	E_{x-max} kN	V_x kN/m	V_{x-max} kN/m
C201	971	8.64	0.43	160.6	118.68	106.81	255.2	332.3	29.5	38.5

$$\begin{aligned} \text{Load per column} &= 29.5 \times 8.64 / 13 \\ &= 19.6 \text{ kN} \end{aligned}$$





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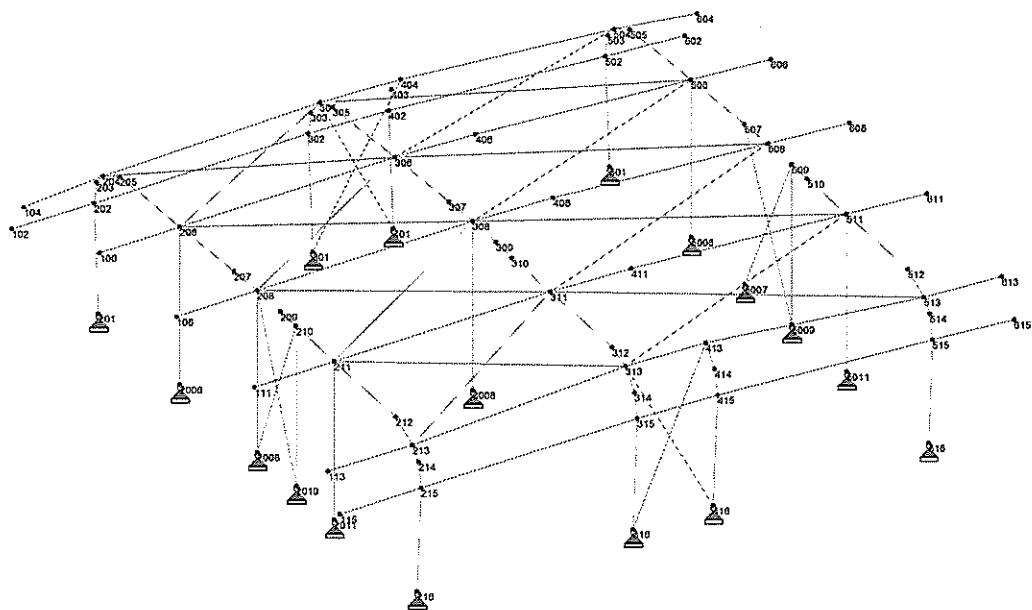
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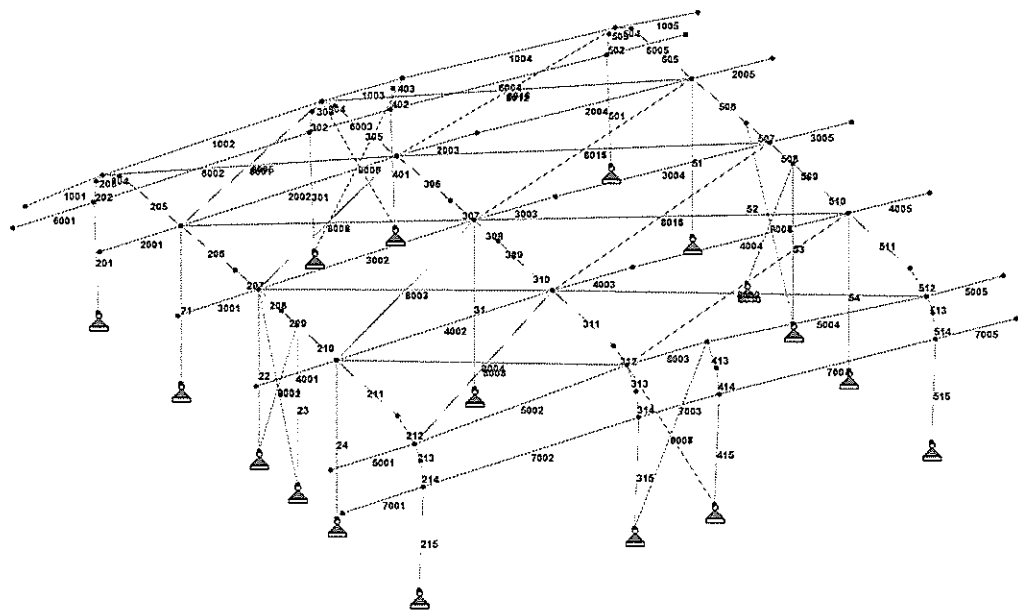
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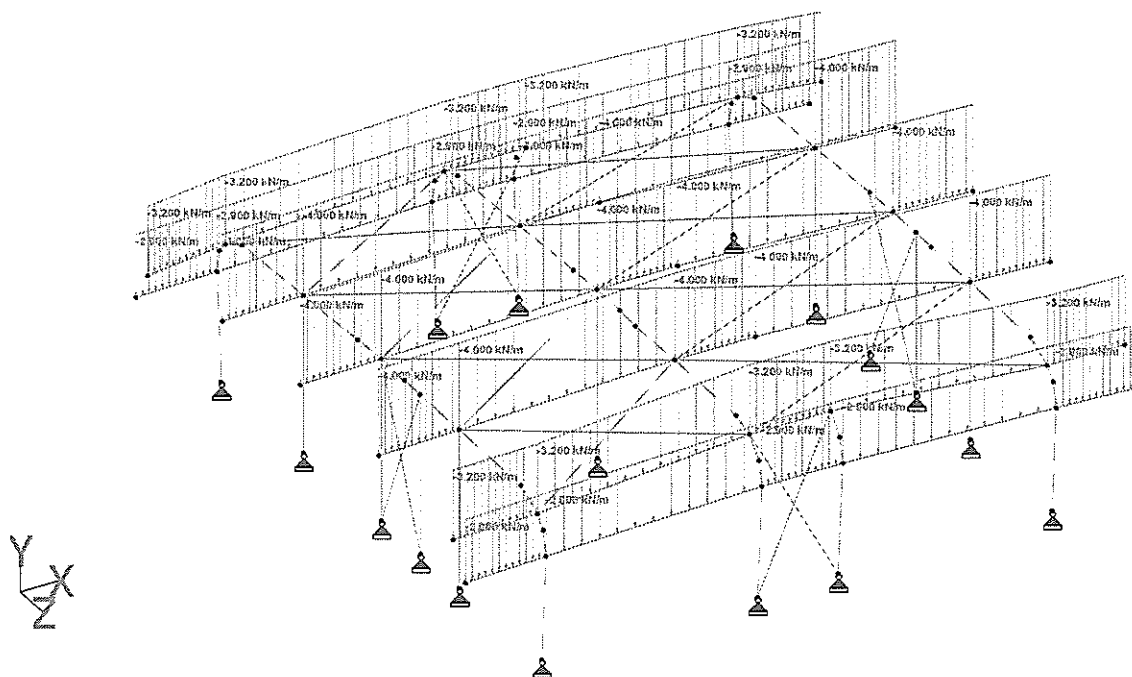


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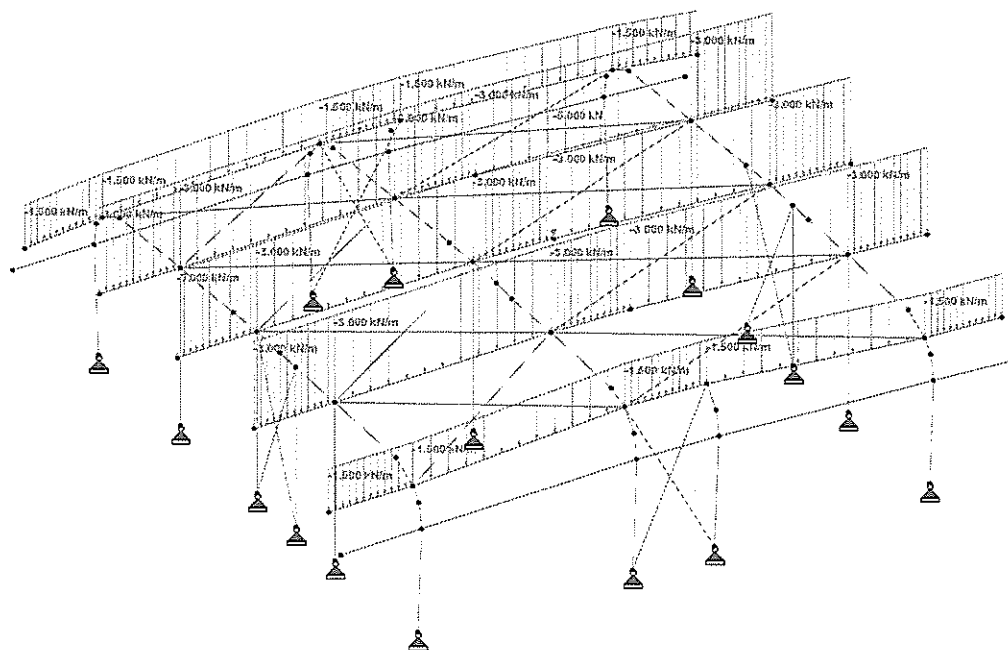


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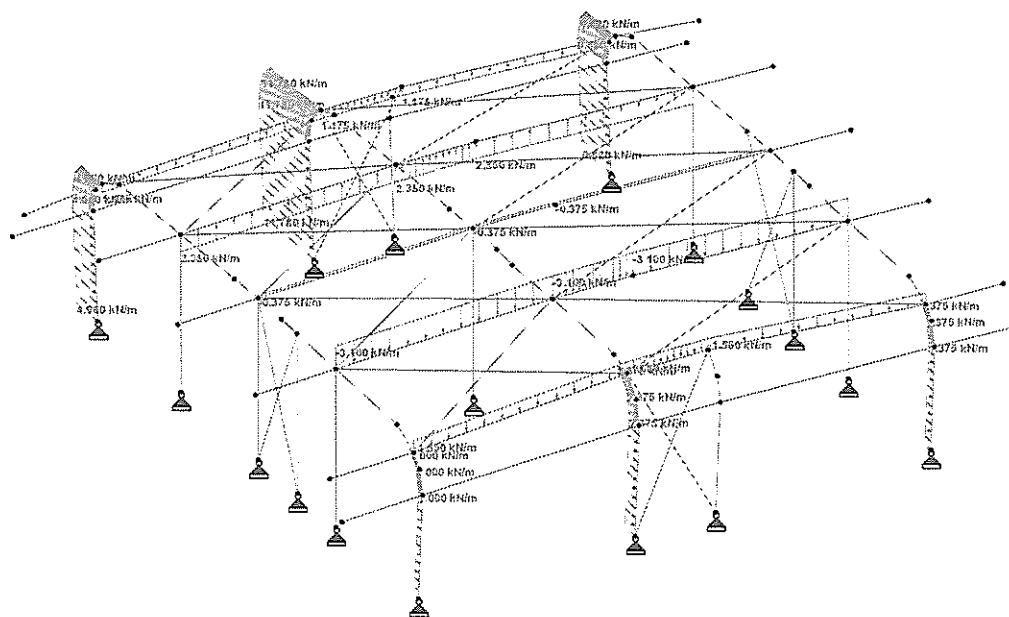


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Load 11



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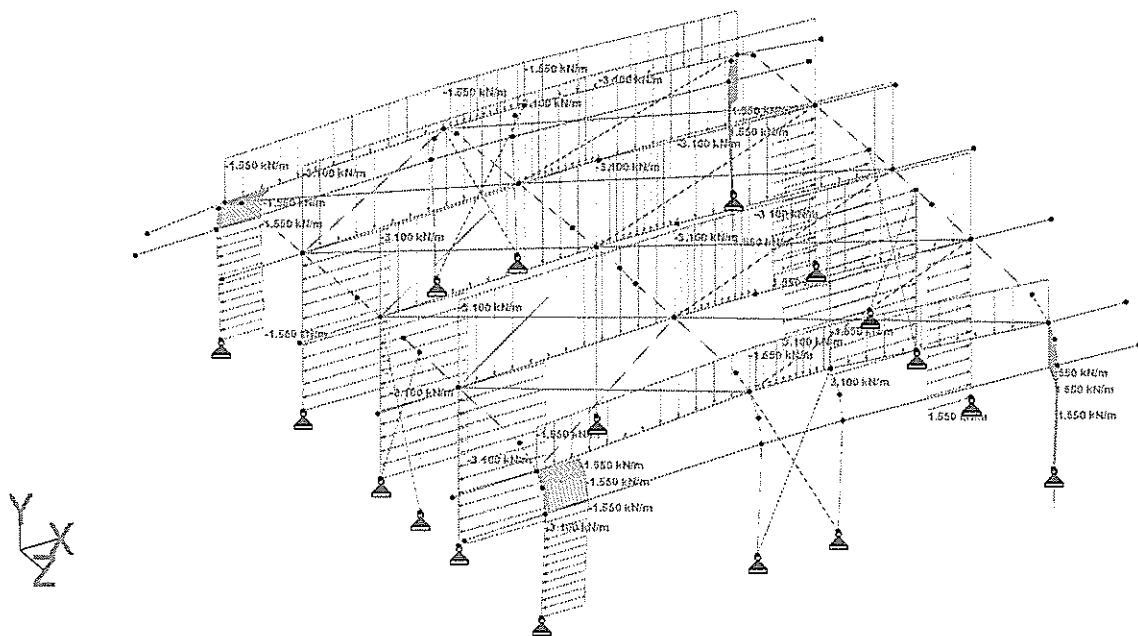
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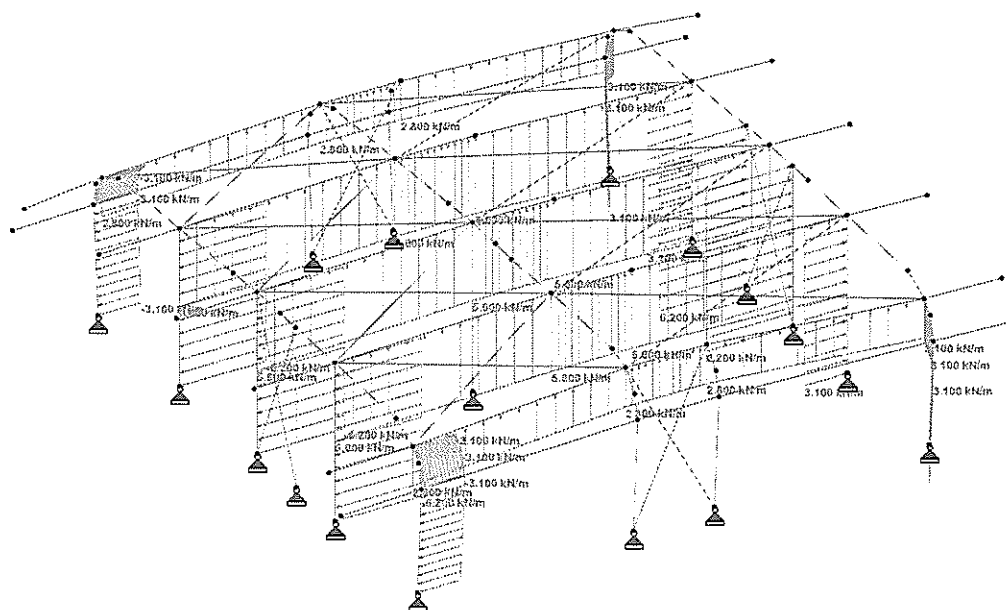
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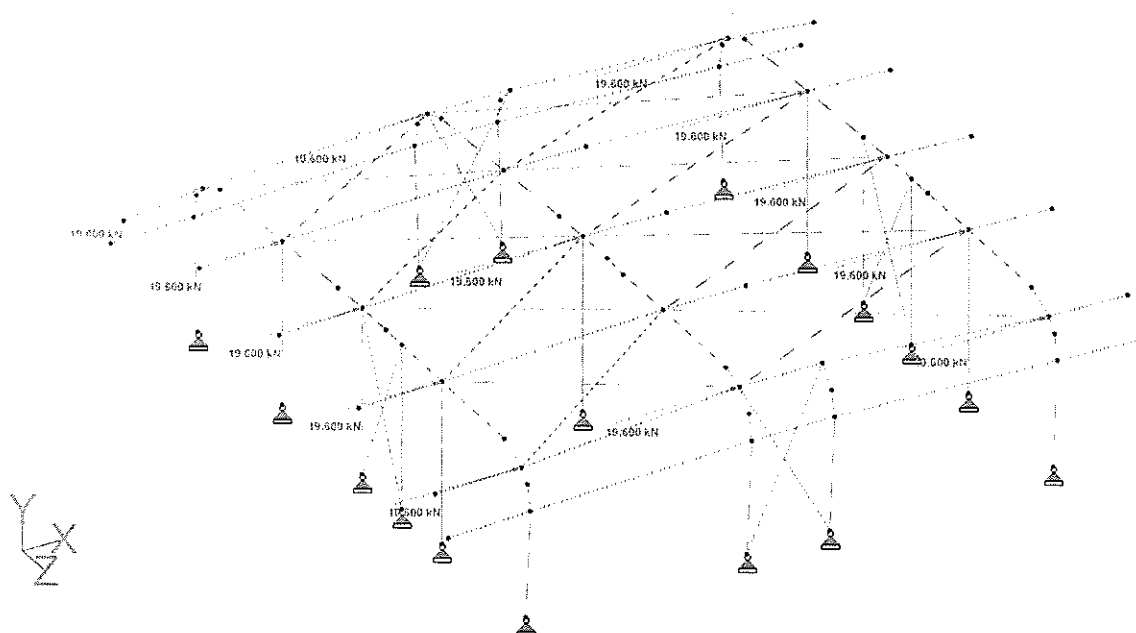
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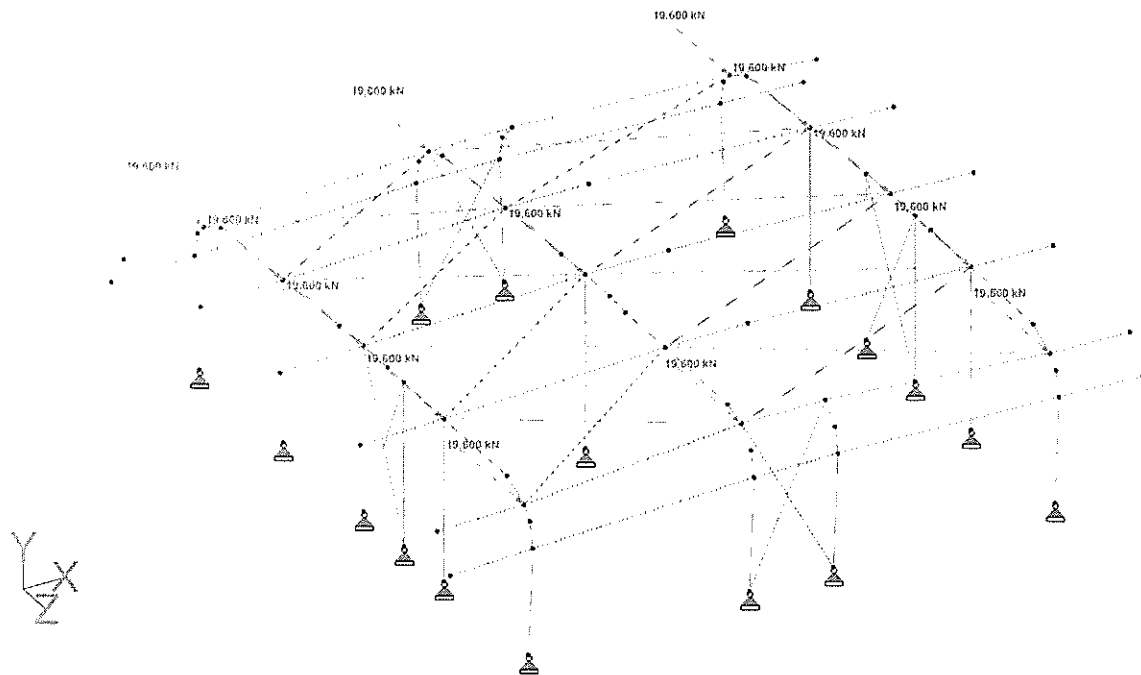


Load 21



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Load 22

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 * Research Engineers, Intl.
 * Date= APR 28, 2007
 * Time= 15: 1: 4
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 * USER ID: Meinhardt Infrastructure Pte Ltd
 * *****

1. STAAD SPACE

INPUT FILE: Vehicle Maintenance Shop-r1.STD
 2. START JOB INFORMATION
 3. ENGINEER DATE APRIL 2007
 4. END JOB INFORMATION
 5. INPUT WIDTH 79
 6. UNIT METER KN

7. JOINT COORDINATES
 8. 102 -12.500 3.860 -10.586; 104 -12.500 5.067 -9.804
 9. 106 -12.500 5.764 -5.000; 108 -12.500 5.891 0.000
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 43. 513 9.500 5.265 10.000; 516 9.500 0.000 10.279
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STAAD SPACE

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 46. 602 12.500 3.860 -10.586; 604 12.500 5.067 -9.804
 47. 606 12.500 5.764 -5.000; 608 12.500 5.891 0.000
 48. 611 12.500 5.764 5.000; 613 12.500 5.067 9.804
 49. 615 12.500 3.860 10.586
 50. 206 -9.500 0.000 -10.279; 207 -9.500 3.965 -10.556
 51. 208 -9.500 0.000 -10.279; 209 -9.500 3.965 -10.556
 52. 210 -9.500 0.000 -10.279; 211 -9.500 3.965 -10.556
 53. 201 -9.500 0.000 2.500; 202 -9.500 0.000 5.000
 54. 308 -1.500 0.000 0.000
 55. 506 9.500 0.000 -5.000; 507 9.500 0.000 -1.500
 56. 509 9.500 0.000 1.500; 511 9.500 0.000 5.000
 57. MEMBER INCIDENCES
 58. *COLUMN
 59. 21 2006 206; 22 2008 208; 23 2010 210; 24 2011 211
 60. 31 3008 308
 61. 51 5006 506; 52 5007 507; 53 5009 509; 54 5011 511
 62. *ROOF BEAM
 63. 201 201 202 215; 301 301 302 315
 64. 401 401 402 403; 413 413 414 415
 65. 501 501 502 515
 66. 1001 1004 204; 1002 304 304; 1003 304 404
 67. 1004 404 504; 1005 504 604
 68. 1001 106 206; 1002 206 306; 1003 306 406
 69. 1004 406 506; 1005 506 606
 70. 1001 108 208; 1002 208 308; 1003 308 408
 71. 1004 408 508; 1005 508 608
 72. 1001 108 208; 1002 208 308; 1003 308 408

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73. 3004 408 508; 3005 508 608
 74. 4001 111 211; 4002 211 311; 4003 311 411
 75. 4004 411 511; 4005 511 611
 76. 5001 113 213; 5002 213 313; 5003 313 413
 77. 5004 413 513; 5005 513 613
 78. *SIDE BEAM
 79. 6001 102 202; 6002 202 302; 6003 302 402
 80. 6004 402 502; 6005 502 602
 81. 7001 115 215; 7002 215 315; 7003 315 415
 82. 7004 415 515; 7005 515 615
 83. *HORIZONTAL BRACING
 84. 8001 204 306; 8002 306 408; 8003 408 511; 8004 511 613
 85. 8005 306 408; 8006 408 511; 8007 511 613
 86. 8009 313 511; 8010 511 613; 8011 613 7003; 8012 7003 8016
 87. 8013 513 613; 8014 613 7003; 8015 7003 8016
 88. 9001 308 2010; 9002 210 2008; 9003 507 5009; 9004 509 5007
 89. 9005 509 5007
 90. 9001 301 404; 9006 401 304; 9007 316 413; 9008 416 313
 91. DEFINE MATERIAL START
 92. ISOTROPIC STEEL
 93. E 2.05E+008
 94. POISSON 0.3
 95. DENSITY 76.8195
 96. ALPHA 1.2E-005
 97. DAMP 0.03
 98. END DEFINE MATERIAL
 100. MEMBER PROPERTY BRITISH
 101. 21 TO 24 31 -
 102. 51 TO 54 TAPERED 0.3 0.006 0.3 0.2 0.01 0.2 0.01
 103. 201 TO 203 213 TO 215 301 TO 303 313 TO 315 -
 104. 401 TO 403 413 TO 415 501 TO 503 -
 105. 513 TO 515 TAPERED 0.3 0.006 0.3 0.2 0.01 0.2 0.01
 106. 204 TO 212 304 TO 312 -
 107. 504 TO 512 TAPERED 0.3 0.006 0.3 0.2 0.01 0.2 0.01
 108. 2001 TO 2005 3001 TO 3005 -
 109. 4001 TO 4005 TAPERED 0.3 0.006 0.3 0.15 0.008 0.15 0.008
 110. 1001 TO 1005 -
 111. 5001 TO 5005 TAPERED 0.3 0.006 0.3 0.15 0.008 0.15 0.008
 112. 6001 TO 6005 TAPERED 0.3 0.010 0.3 0.25 0.015 0.25 0.015
 113. 7001 TO 7005 TAPERED 0.3 0.010 0.3 0.25 0.015 0.25 0.015
 114. 8001 TO 8016 9001 TO 9008 TABLE 5T UAG90X90X6
 115. CONSTANTS
 116. BETA 90 MEMB 21 TO 24 31 TO 54 6001 TO 6005 7001 TO 7005
 117. MATERIAL STEEL ALL
 118. MEMBER TENSION
 119. 8001 TO 8016 9001 TO 9008
 120. 1001 TO 1011 2011 TO 2011 3008 5006 5007 5009 5011 PINNED
 121. 201 215 301 316 401 416 501 516 -
 122. 2006 2008 2010 2011 3008 5006 5007 5009 5011 PINNED
 123. LOAD 1 SELFWEIGHT
 124. LOAD 2 SDF
 125. LOAD 3 SDF
 126. LOAD 4 SDF
 127. MEMBER LOAD
 128. 1001 TO 1005 5001 TO 5005 UNI GY -3.2
 129. 2001 TO 2005 3001 TO 3005 UNI GY -4
 130. 6001 TO 6005 7001 TO 7005 UNI GY -2.9
 131. LOAD 3 LL
 132. MEMBER LOAD
 133. 1001 TO 1005 5001 TO 5005 UNI GY -1.5
 134. 2001 TO 2005 3001 TO 3005 4001 TO 4005 UNI GY -3
 135. CORNER LOAD
 136. *WIND LOAD
 137. 408 FY -5
 138. *WIND LOAD
 139. *ROOF WIND LOAD WILL BE TRANSFERRED THROUGH ROOF SECONDARY BEAM
 140. *AND WALL WIND LOAD WILL BE ACTING ON THE MAIN FRAME
 141. LOAD 11 WL1 (INTERNAL SUCTION, WIND ANGLE 0)
 142. MEMBER LOAD
 143. 1002 TO 1004 UNI GY 1.175
 144. 2002 TO 2004 UNI GY 2.35
 145. 3002 TO 3004 UNI GY -0.375
 146. 4002 TO 4004 UNI GY -3.1
 147. 5002 TO 5004 UNI GY -1.55
 148. 201 TO 203 UNI GZ 4.96
 149. 301 TO 303 UNI GZ 11.78
 150. 501 TO 503 UNI GZ 6.82
 151. 601 TO 603 UNI GZ 1.05
 152. 701 TO 703 UNI GZ 2.975
 153. 801 TO 803 UNI GZ 1.375
 154. 901 TO 903 UNI GZ 1.375
 155. LOAD 12 WL2 (INTERNAL PRESSURE, WIND ANGLE 0)
 156. MEMBER LOAD
 157. 1002 TO 1004 UNI GY 2.725
 158. 2002 TO 2004 UNI GY 5.45
 159. 3002 TO 3004 UNI GY 3.975
 160. 4002 TO 4004 UNI GY 2.5

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Vehicle Maintenance Shop-r1-ANL									
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z	-- PAGE NO. 10	
STAAD SPACE									
SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									

1212	-0.96	43.18	-14.91	0.00	0.00	0.00	0.00		
1013	1.04	70.53	-1.01	0.00	0.00	0.00	0.00		
1113	0.95	68.60	-1.35	0.00	0.00	0.00	0.00		
1213	0.98	47.59	-0.82	0.00	0.00	0.00	0.00		
1014	0.20	45.34	-1.08	0.00	0.00	0.00	0.00		
1114	-0.03	39.21	-1.42	0.00	0.00	0.00	0.00		
1214	0.00	18.20	-0.89	0.00	0.00	0.00	0.00		
1021	-3.22	72.15	-1.46	0.00	0.00	0.00	0.00		
1121	-0.04	64.04	-5.83	0.00	0.00	0.00	0.00		
1221	3.22	63.70	-1.61	0.00	0.00	0.00	0.00		
1122	0.03	71.81	2.76	0.00	0.00	0.00	0.00		
1222	-3.29	51.50	-1.12	0.00	0.00	0.00	0.00		
STAAD SPACE									
SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									

1222	-0.11	43.39	-5.49	0.00	0.00	0.00	0.00		
1321	3.15	43.05	-1.27	0.00	0.00	0.00	0.00		
1322	-0.04	51.16	3.10	0.00	0.00	0.00	0.00		
1001	0.19	89.18	1.67	0.00	0.00	0.00	0.00		
1011	0.07	82.00	-4.06	0.00	0.00	0.00	0.00		
1111	-0.18	81.98	-4.37	0.00	0.00	0.00	0.00		
1211	-0.15	60.97	-5.10	0.00	0.00	0.00	0.00		
1012	-0.01	67.65	-13.68	0.00	0.00	0.00	0.00		
1112	-0.28	65.24	-15.79	0.00	0.00	0.00	0.00		
1212	-0.25	44.22	-16.32	0.00	0.00	0.00	0.00		
1013	1.05	70.53	1.01	0.00	0.00	0.00	0.00		
1113	0.95	68.60	1.35	0.00	0.00	0.00	0.00		
1213	0.98	47.59	0.82	0.00	0.00	0.00	0.00		
1014	0.20	45.36	1.06	0.00	0.00	0.00	0.00		
1114	-0.03	39.23	1.41	0.00	0.00	0.00	0.00		
1214	0.00	18.22	0.88	0.00	0.00	0.00	0.00		
1021	-3.22	72.15	1.45	0.00	0.00	0.00	0.00		
1121	-0.16	71.93	-2.51	0.00	0.00	0.00	0.00		
1221	3.22	63.69	1.61	0.00	0.00	0.00	0.00		
1122	0.17	63.92	5.57	0.00	0.00	0.00	0.00		
1222	-3.29	51.51	1.11	0.00	0.00	0.00	0.00		
1321	-0.24	51.28	-2.85	0.00	0.00	0.00	0.00		
1322	3.15	43.05	1.27	0.00	0.00	0.00	0.00		
1001	0.09	43.27	5.23	0.00	0.00	0.00	0.00		
1011	-0.08	89.56	-0.51	0.00	0.00	0.00	0.00		
1111	-0.52	56.80	-1.67	0.00	0.00	0.00	0.00		
1211	-0.01	55.75	-1.85	0.00	0.00	0.00	0.00		
1012	-0.74	49.14	-1.72	0.00	0.00	0.00	0.00		
1112	-0.29	47.14	-1.93	0.00	0.00	0.00	0.00		
1212	-0.24	24.46	-1.91	0.00	0.00	0.00	0.00		
1013	1.50	8.84	1.80	0.00	0.00	0.00	0.00		
1113	10.91	83.65	-0.48	0.00	0.00	0.00	0.00		
1213	12.29	67.05	-0.45	0.00	0.00	0.00	0.00		
1014	12.08	51.43	-0.34	0.00	0.00	0.00	0.00		
1114	17.03	44.92	-0.27	0.00	0.00	0.00	0.00		
1214	19.43	21.86	-0.20	0.00	0.00	0.00	0.00		
1021	15.22	6.25	-0.09	0.00	0.00	0.00	0.00		
1121	-0.44	55.69	-0.24	0.00	0.00	0.00	0.00		
1221	0.84	58.21	-5.22	0.00	0.00	0.00	0.00		
1121	2.01	59.82	-0.49	0.00	0.00	0.00	0.00		
1222	-0.72	37.29	4.49	0.00	0.00	0.00	0.00		
1122	-0.25	35.67	-0.12	0.00	0.00	0.00	0.00		
1221	0.43	37.20	-0.30	0.00	0.00	0.00	0.00		
1321	1.70	37.20	-0.37	0.00	0.00	0.00	0.00		
1322	0.41	34.68	4.63	0.00	0.00	0.00	0.00		
2008	1001	-0.37	73.00	-0.11	0.00	0.00	0.00		
1011	-0.37	-17.88	-35.01	0.00	0.00	0.00	0.00		
1111	-0.22	-45.74	-40.81	0.00	0.00	0.00	0.00		
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Vehicle Maintenance Shop-r1.ANL									
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z		
5006	1321	1.95	91.08	0.03	0.00	0.00	0.00		
	1322	0.71	3.97	0.00	0.00	0.00	0.00		
	1001	-4.12	100.91	-0.81	0.00	0.00	0.00		
	1011	-1.55	57.19	-1.94	0.00	0.00	0.00		
	1111	-0.24	31.80	-2.03	0.00	0.00	0.00		
	1211	0.42	14.37	-1.87	0.00	0.00	0.00		
	1012	0.77	43.37	-2.10	0.00	0.00	0.00		
	1112	2.11	15.97	-2.22	0.00	0.00	0.00		
	1212	6.78	11.75	-2.97	0.00	0.00	0.00		
	1013	-6.78	89.08	-0.16	0.00	0.00	0.00		
	1113	-0.12	83.60	-0.50	0.00	0.00	0.00		
	1213	-0.12	83.60	-0.50	0.00	0.00	0.00		
	1014	-8.33	46.32	-0.41	0.00	0.00	0.00		
	1114	-8.33	19.12	-0.25	0.00	0.00	0.00		
	1214	-7.49	1.69	-0.10	0.00	0.00	0.00		
	1021	-3.75	66.29	-0.58	0.00	0.00	0.00		
	1022	-2.61	61.41	-4.74	0.00	0.00	0.00		
	STAAD SPACE								

Vehicle Maintenance Shop-r1.ANL

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z		
5007	1121	-1.36	63.23	-0.53	0.00	0.00	0.00		
	1122	-2.50	68.11	-3.62	0.00	0.00	0.00		
	1222	-2.69	40.75	-0.38	0.00	0.00	0.00		
	1322	-1.55	35.87	-4.54	0.00	0.00	0.00		
	1011	-0.30	37.68	-0.33	0.00	0.00	0.00		
	1111	-1.44	42.57	3.82	0.00	0.00	0.00		
	1211	-0.03	55.28	1.38	0.00	0.00	0.00		
	1012	-0.03	-37.40	-44.35	0.00	0.00	0.00		
	1112	-0.02	-62.94	-52.26	0.00	0.00	0.00		
	1212	-0.01	-72.43	-52.48	0.00	0.00	0.00		
	1013	0.00	-54.35	-46.56	0.00	0.00	0.00		
	1113	0.01	-82.71	-54.84	0.00	0.00	0.00		
	1213	0.02	-82.71	-55.97	0.00	0.00	0.00		
	1014	-1.36	43.45	1.82	0.00	0.00	0.00		
	1114	-1.36	33.96	0.83	0.00	0.00	0.00		
	1214	-2.50	27.41	-0.32	0.00	0.00	0.00		
	1021	-2.50	12.67	-0.89	0.00	0.00	0.00		
	1121	-0.03	3.18	-1.12	0.00	0.00	0.00		
	1221	-0.02	-158.81	-102.25	0.00	0.00	0.00		
	1022	-0.02	35.08	0.77	0.00	0.00	0.00		
	1122	-0.02	229.54	103.96	0.00	0.00	0.00		
	1222	-0.02	21.64	0.94	0.00	0.00	0.00		
	1023	-0.01	-172.82	-102.61	0.00	0.00	0.00		
	1123	0.00	21.06	0.59	0.00	0.00	0.00		
	1223	-0.01	215.53	103.61	0.00	0.00	0.00		
	1024	-0.03	55.28	-1.38	0.00	0.00	0.00		
	1124	-0.04	150.00	-2.86	0.00	0.00	0.00		
	1224	-0.03	132.87	-2.83	0.00	0.00	0.00		
	1025	-0.01	119.97	-2.63	0.00	0.00	0.00		
	1125	0.01	120.66	-2.54	0.00	0.00	0.00		
	1225	0.01	111.17	-2.32	0.00	0.00	0.00		
	1026	-11.11	53.79	-1.35	0.00	0.00	0.00		
	1126	-12.95	43.45	-1.05	0.00	0.00	0.00		
	1226	-12.94	33.96	-0.83	0.00	0.00	0.00		
	1027	-22.13	28.16	0.11	0.00	0.00	0.00		
	1127	-25.80	13.55	0.66	0.00	0.00	0.00		
	1227	-25.79	4.06	0.88	0.00	0.00	0.00		
	1028	-0.03	35.74	-0.79	0.00	0.00	0.00		
	1128	-0.02	227.47	-4.79	0.00	0.00	0.00		
	1228	-0.01	35.00	-0.92	0.00	0.00	0.00		
	1029	-0.02	-136.73	-3.08	0.00	0.00	0.00		
	1129	-0.02	21.73	-0.44	0.00	0.00	0.00		
	1229	-0.01	213.45	-4.44	0.00	0.00	0.00		
	1030	0.00	20.75	-0.53	0.00	0.00	0.00		
	1130	-0.01	-170.75	3.43	0.00	0.00	0.00		
	STAAD SPACE								

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z		
5009	1001	4.62	0.00	0.00	0.00	0.00	0.00		
	1011	19.43	0.00	0.00	0.00	0.00	0.00		
	1111	-5.22	5.87	0.00	0.00	0.00	0.00		
	1211	-31.75	5.87	0.00	0.00	0.00	0.00		
	1012	4.62	0.00	0.00	0.00	0.00	0.00		
	1112	19.43	0.00	0.00	0.00	0.00	0.00		
	1212	-5.22	5.87	0.00	0.00	0.00	0.00		
	1013	4.62	0.00	0.00	0.00	0.00	0.00		
	1113	19.43	0.00	0.00	0.00	0.00	0.00		
	1213	-5.22	5.87	0.00	0.00	0.00	0.00		
	1014	4.62	0.00	0.00	0.00	0.00	0.00		
	1114	19.43	0.00	0.00	0.00	0.00	0.00		
	1214	-5.22	5.87	0.00	0.00	0.00	0.00		
	1021	4.62	0.00	0.00	0.00	0.00	0.00		
	1121	19.43	0.00	0.00	0.00	0.00	0.00		
	1221	-5.22	5.87	0.00	0.00	0.00	0.00		
	1022	4.62	0.00	0.00	0.00	0.00	0.00		
	1122	19.43	0.00	0.00	0.00	0.00	0.00		
	1222	-5.22	5.87	0.00	0.00	0.00	0.00		
	1023	4.62	0.00	0.00	0.00	0.00	0.00		
	1123	19.43	0.00	0.00	0.00	0.00	0.00		
	1223	-5.22	5.87	0.00	0.00	0.00	0.00		
	1024	4.62	0.00	0.00	0.00	0.00	0.00		
	1124	19.43	0.00	0.00	0.00	0.00	0.00		
	1224	-5.22	5.87	0.00	0.00	0.00	0.00		
	1025	4.62	0.00	0.00	0.00	0.00	0.00		
	1125	19.43	0.00	0.00	0.00	0.00	0.00		
	1225	-5.22	5.87	0.00	0.00	0.00	0.00		
	1026	4.62	0.00	0.00	0.00	0.00	0.00		
	1126	19.43	0.00	0.00	0.00	0.00	0.00		
	1226	-5.22	5.87	0.00	0.00	0.00	0.00		
	1027	4.62	0.00	0.00	0.00	0.00	0.00		
	1127	19.43	0.00	0.00	0.00	0.00	0.00		
	1227	-5.22	5.87	0.00	0.00	0.00	0.00		
	1028	4.62	0.00	0.00	0.00	0.00	0.00		
	1128	19.43	0.00	0.00	0.00	0.00	0.00		
	1228	-5.22	5.87	0.00	0.00	0.00	0.00		
	1029	4.62	0.00	0.00	0.00	0.00	0.00		
	1129	19.43	0.00	0.00	0.00	0.00	0.00		
	1229	-5.22	5.87	0.00	0.00	0.00	0.00		
	1030	4.62	0.00	0.00	0.00	0.00	0.00		
	1130	19.43	0.00	0.00	0.00	0.00	0.00		
	1230	-5.22	5.87	0.00	0.00	0.00	0.00		
	STAAD SPACE								

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

Vehicle Maintenance Shop-r1.ANL									
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z		
5011	1001	-4.13	100.92	0.81	0.00	0.00	0.00		
	1011	-5.79	105.93	-1.18	0.00	0.00	0.00		
	1111	-5.18	88.66	-1.60	0.00	0.00	0.00		
	1211	-4.51	71.23	-1.76	0.00	0.00	0.00		
	1012	-1.81	76.56	-1.12	0.00	0.00	0.00		
	1112	-0.54	54.39	-1.54	0.00	0.00	0.00		
	1212	0.13	38.96	-1.92	0.00	0.00	0.00		
	1013	-6.78	89.08	-0.16	0.00	0.00	0.00		
	1113	-0.12	83.60	-0.50	0.00	0.00	0.00		
	1213	-0.12	83.60	-0.50	0.00	0.00	0.00		
	1014	-8.33	46.32	-0.41	0.00	0.00	0.00		
	1114	-8.33	19.14	-0.24	0.00	0.00	0.00		
	1214	-7.49	1.71	-0.08	0.00	0.00	0.00		
	1021	-3.76	66.30	-0.58	0.00	0.00	0.00		
	1121	-1.36	63.23	-0.53	0.00	0.00	0.00		
	1221	-2.58	58.77	-0.38	0.00	0.00	0.00		
	1022	-1.48	45.21	-4.12	0.00	0.00	0.00		
	1122	-1.48	45.21	-4.12	0.00	0.00	0.00		
	1222	-0.30	37.68	-0.33	0.00	0.00	0.00		
	1023	-1.52	33.22	-4.83	0.00	0.00	0.00		
	STAAD SPACE								

Vehicle Maintenance Shop-r1.ANL

266. PRINT FORCE ENVELOPE NSECTION 5 ALL

FORCE ENVELOPE NSECTION 5

STAAD SPACE

MEMBER FORCE ENVELOPE

ALL UNITS ARE KN METE

MEMB DISTANCE FY LD MZ LD FZ LD MY LD

21 0.00 MAX 4.62 1322 0.00 1022 19.43 1114 0.00 1021

MIN -5.22 1022 -6.13 1022 1.50 1212 16.83 1114

1.17 MAX 4.62 1322 6.13 1022 1.50 1212 16.83 1114

2.35 MIN -5.22 1022 -10.84 1022 2.17 1013 21.69 1114

3.52 MAX 4.62 1322 18.39 1022 2.01 1121 16.35 1113

206, AMONGST ALL SECT LOCATIONS												
MAX/MIN FORCE VALUES FOR MEMB			MZ/ DIST LD			MZ/ DIST LD			MZ/ DIST LD			
FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	
3.50 MAX	6.69 1122		6.33 1222			6.33 1222			33.34 C	0.00 1222		
MIN	-5.76 1222		-7.48 1122			-7.48 1122			50.74 T	3.50 1122		
STAAD SPACE												
207, AMONGST ALL SECT LOCATIONS												
MAX/MIN FORCE VALUES FOR MEMB			MZ/ DIST LD			MZ/ DIST LD			MZ/ DIST LD			
FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	
6.68 MAX	0.00 1122		15.41 1.50 1222			15.41 1.50 1222			33.31 C	0.00 1222		
0.91 MIN	0.00 1014		-0.91 1.50 1022			-0.91 1.50 1022			50.76 T	1.50 1122		
-6.34 MAX	1.50 1222		-0.38 1.50 1321			-0.38 1.50 1321						
-0.18 MIN	1.50 1321											
STAAD SPACE												
208, AMONGST ALL SECT LOCATIONS												
MAX/MIN FORCE VALUES FOR MEMB			MZ/ DIST LD			MZ/ DIST LD			MZ/ DIST LD			
FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	
4.67 MAX	0.00 1322		8.54 1322			8.54 1322			0.58 1021			
0.18 MIN	0.00 1321		-10.67 1022			-10.67 1022			-0.37 1321			
-6.54 MAX	1.50 1022		7.16 1322			7.16 1322			0.54 1021			
-0.15 MIN	1.50 1014		-8.92 1022			-8.92 1022			-0.32 1321			
STAAD SPACE												
209, AMONGST ALL SECT LOCATIONS												
MAX/MIN FORCE VALUES FOR MEMB			MZ/ DIST LD			MZ/ DIST LD			MZ/ DIST LD			
FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	
4.67 MAX	0.00 1322		8.54 1322			8.54 1322			0.58 1021			
0.18 MIN	0.00 1321		-10.67 1022			-10.67 1022			-0.37 1321			
-6.54 MAX	1.50 1022		7.16 1322			7.16 1322			0.54 1021			
-0.15 MIN	1.50 1014		-8.92 1022			-8.92 1022			-0.32 1321			
STAAD SPACE												

[illegible]

Vehicle Maintenance Shop-F1-ANL									
515	0.00	MAX	7.32	1212	12.80	1212	12.80	1212	12.80
	0.79	MIN	-7.32	1212	-12.80	1212	-12.80	1212	-12.80
	1.59	MAX	7.32	1212	12.80	1212	12.80	1212	12.80
	2.38	MIN	-7.32	1212	-12.80	1212	-12.80	1212	-12.80
	3.18	MAX	7.32	1212	12.80	1212	12.80	1212	12.80
	3.97	MIN	-7.32	1212	-12.80	1212	-12.80	1212	-12.80
	STAAD SPACE								

Vehicle Maintenance Shop-F1-ANL									
515	0.00	MAX	7.32	1212	12.80	1212	12.80	1212	12.80
	0.79	MIN	-7.32	1212	-12.80	1212	-12.80	1212	-12.80
	1.59	MAX	7.32	1212	12.80	1212	12.80	1212	12.80
	2.38	MIN	-7.32	1212	-12.80	1212	-12.80	1212	-12.80
	3.18	MAX	7.32	1212	12.80	1212	12.80	1212	12.80
	3.97	MIN	-7.32	1212	-12.80	1212	-12.80	1212	-12.80
	STAAD SPACE								

MAX/MIN FORCE VALUES FOR MEMB 515, AMONGST ALL SECT LOCATIONS									
	FZ	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	7.32	0.00	1212	8.99	0.00	1322	89.08	C	3.97 1001
MIN.	-20.30	3.97	1112	-30.54	0.79	1112	18.24	C	3.97 1214
	-3.22	3.97	1112	-13.08	0.00	1221			

MAX/MIN FORCE VALUES FOR MEMB 515, AMONGST ALL SECT LOCATIONS									
	FZ	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	7.32	0.00	1212	8.99	0.00	1322	89.08	C	3.97 1001
MIN.	-20.30	3.97	1112	-30.54	0.79	1112	18.24	C	3.97 1214
	-3.22	3.97	1112	-13.08	0.00	1221			

MAX/MIN FORCE VALUES FOR MEMB 1001, AMONGST ALL SECT LOCATIONS									
	FZ	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	1321	0.00	1222	0.00	1121	0.00	1221
MIN.	-20.00	0.00	1321	-1.33	1001	0.00	1121	0.00	1121
	-20.00	0.00	1321	-1.33	1001	0.00	1121	0.00	1121

MAX/MIN FORCE VALUES FOR MEMB 1001, AMONGST ALL SECT LOCATIONS									
	FZ	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	1321	0.00	1222	0.00	1121	0.00	1221
MIN.	-20.00	0.00	1321	-1.33	1001	0.00	1121	0.00	1121
	-20.00	0.00	1321	-1.33	1001	0.00	1121	0.00	1121

MAX/MIN FORCE VALUES FOR MEMB 1002, AMONGST ALL SECT LOCATIONS									
	FZ	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	1013	35.54	0.00	1001	0.16	1013	1.22 1214
MIN.	-20.00	0.00	1013	-14.00	0.00	1013	-0.29	1214	-1.87 1013
	-20.00	0.00	1013	-14.00	0.00	1013	-0.29	1214	-1.87 1013

MAX/MIN FORCE VALUES FOR MEMB 1002, AMONGST ALL SECT LOCATIONS									
	FZ	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	1013	35.54	0.00	1001	0.16	1013	1.22 1214
MIN.	-20.00	0.00	1013	-14.00	0.00	1013	-0.29	1214	-1.87 1013
	-20.00	0.00	1013	-14.00	0.00	1013	-0.29	1214	-1.87 1013

MAX/MIN FORCE VALUES FOR MEMB 1003, AMONGST ALL SECT LOCATIONS									
	FZ	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	1013	35.54	0.00	1001	0.16	1013	1.22 1214
MIN.	-20.00	0.00	1013	-14.00	0.00	1013	-0.29	1214	-1.87 1013
	-20.00	0.00	1013	-14.00	0.00	1013	-0.29	1214	-1.87 1013

MAX/MIN FORCE VALUES FOR MEMB 1003, AMONGST ALL SECT LOCATIONS									
	FZ	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	1013	35.54	0.00	1001	0.16	1013	1.22 1214
MIN.	-20.00	0.00	1013	-14.00	0.00	1013	-0.29	1214	-1.87 1013
	-20.00	0.00	1013	-14.00	0.00	1013	-0.29	1214	-1.87 1013

MAX.	37.03	0.00	131.51	8.01	1013	18.61	C	0.00	1021
	0.03	0.00	0.14	8.01	1111				
MIN.	-62.96	8.01	-40.12	8.01	1214				
	-0.02	8.01	-0.08	0.00	1122	19.85	T	0.00	1214

MAX/MIN		FORCE VALUES FOR MEMB				3003, AMONGST ALL SECT LOCATIONS					
		F _Y	F _Z	DIST	LD	NZ/	DIST	LD	FX	DIST	LD
3.00	MAX	45.28	1013	12.38	1214				0.02	1022	0.05 1321
	MIN	-14.00	1214	-45.55	1013				-0.02	1322	-0.06 1112

STAAD SPACE

[illegible]

MAX.	39.39	0.00 1013	49.15	8.01 1001		25.96 C	8.01 1022
	0.02	0.00 1022	0.12	8.01 1022			
MIN.	-60.59	8.01 1013	-107.66	3.20 1013			
	-0.02	8.01 1322	-0.12	8.01 1322		22.02 T	8.01 1214
3005	0.00	32.52 1001	48.82 1001	0.00 1222		0.00 1122	0.00 1122
	MIN	11.65 1322	17.49 1322	0.00 1122		0.00 1222	0.00 1222
	0.60	26.02 1001	31.24 1001	0.00 1222		0.00 1212	0.00 1212
	MIN	9.32 1322	11.19 1322	0.00 1122		0.00 1321	0.00 1321
	1.20	19.51 1001	17.57 1001	0.00 1222		0.00 1021	0.00 1021
	MIN	6.99 1322	6.30 1322	0.00 1122		0.00 1321	0.00 1321
	1.80	13.01 1001	7.81 1001	0.00 1222		0.00 1221	0.00 1221

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Vehicle Maintenance Shop-Pl-AWL												
			MAX/MIN FORCE VALUES FOR MEMB			4004, AMONGST ALL SECT LOCATIONS						
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	DIST LD	FX	DIST LD				
1.60	MAX	15.02	1013	28.56	1214	0.07	1214	0.18	1013			
	MIN	-5.02	1214	-95.27	1011	-0.05	1121	-0.93	1013			
3.20	MAX	8.51	1001	-0.81	1001	-0.05	1121	-0.05	1121			
	MIN	-6.56	1001	-103.26	1013	-0.05	1121	-0.05	1121			
4.80	MAX	6.76	1214	26.57	1214	0.07	1214	0.28	1214			
	MIN	-26.38	1011	-79.23	1013	-0.05	1121	-0.05	1322			
6.40	MAX	11.90	1214	12.03	1214	0.07	1214	0.39	1214			
	MIN	-46.38	1011	-24.38	1113	-0.05	1121	-0.10	1322			
8.01	MAX	17.54	1214	73.67	1011	0.07	1214	0.49	1214			
	MIN	-66.37	1011	-11.54	1214	-0.05	1121	-0.17	1122			
STAAD SPACE												
4005												
	MAX	0.00	MAX	32.52	1001	48.82	1001	0.00	1022	0.00	1322	
	MIN	0.60	MIN	11.65	1322	17.49	1322	0.00	1322	0.00	1022	
				26.02	1001	31.24	1001	0.00	1022	0.00	1322	
				9.32	1322	11.19	1322	0.00	1322	0.00	1022	
	1.20	MAX	19.51	1001	17.57	1001	0.00	1022	0.00	1322	0.00	1321
		MIN	6.99	1322	6.30	1322	0.00	1322	0.00	1021	0.00	1321
	1.80	MAX	13.01	1001	7.81	1001	0.00	1322	0.00	1322	0.00	1321
		MIN	4.66	1322	2.80	1322	0.00	1322	0.00	1022	0.00	1321
	2.40	MAX	6.50	1001	1.95	1001	0.00	1022	0.00	1322	0.00	1022
		MIN	2.33	1322	0.70	1322	0.00	1322	0.00	1022	0.00	1322
	3.00	MAX	0.00	1014	0.00	1011	0.00	1022	0.00	1022	0.00	1022
		MIN	0.00	1321	0.00	1213	0.00	1322	0.00	1322	0.00	1322
-- PAGE NO. 35												
4005, AMONGST ALL SECT LOCATIONS												
	MAX	32.52	0.00	1001	48.82	0.00	1001	0.00	1322	0.00	1021	
	MIN	0.00	0.00	1022	0.00	0.00	1322	0.00	C	3.00	1021	
		0.00	3.00	1321	0.00	3.00	1313					
		0.00	3.00	1322	0.00	0.00	1022	1.17	T	0.00	1001	
5001												
	0.00	MAX	0.00	1222	0.00	1212	0.00	1221	0.00	1121		
	0.60	MIN	0.00	1122	0.00	1322	0.00	1121	0.00	1221		
			-1.90	1322	1.25	1321	0.00	1221	0.00	1113		
	1.21	MAX	-3.80	1322	5.31	1001	0.00	1121	0.00	1221		
		MIN	-8.00	1001	2.29	1322	0.00	1121	0.00	1113		
	1.81	MAX	-5.71	1322	11.94	1001	0.00	1221	0.00	1221		
		MIN	-13.21	1001	5.16	1322	0.00	1121	0.00	1113		
	2.41	MAX	-7.61	1322	21.22	1001	0.00	1221	0.00	1221		
		MIN	-17.61	1001	9.17	1322	0.00	1121	0.00	1113		
	3.01	MAX	-9.51	1322	33.16	1001	0.00	1221	0.00	1221		
		MIN	-22.01	1001	14.33	1322	0.00	1121	0.00	1113		
5001, AMONGST ALL SECT LOCATIONS												
	MAX	0.00	0.00	1222	33.16	3.01	1001	0.00	C	0.00	1013	
	MIN	0.00	0.00	1221	0.00	3.01	1321	1.45	T	3.01	1001	
		-22.01	9.01	1001	0.00	3.00	1322					
		0.00	3.01	1321	0.00	3.01	1113					
5002												
	0.00	MAX	32.50	1011	35.54	1001	0.29	1214	2.09	1011		
	1.60	MIN	0.77	1214	13.99	1214	-0.19	1011	-1.23	1214		
			19.89	1011	12.24	1214	0.29	1214	1.78	1011		
	3.21	MAX	1.42	1214	-11.84	1111	-0.19	1011	-0.77	1214		
		MIN	7.59	1001	9.45	1214	0.29	1214	1.47	1011		
	4.81	MAX	0.95	1221	-33.32	1011	-0.19	1011	-0.30	1214		
		MIN	2.71	1214	5.62	1214	0.29	1214	1.38	1121		
			-5.53	1013	-34.89	1011	-0.19	1011	-0.21	1221		
Page 35												

Vehicle Maintenance Shop-r1-ANL											
6.42	MAX	3.36	1214	8.56	1221	0.29	1214	1.38	1121		
	MIN	-18.14	1013	-2.38	1121	-0.19	1011	-0.40	1221		
8.02	MAX	4.01	1214	32.58	1021	0.29	1214	1.37	1121		
	MIN	-30.75	1013	-7.44	1321	-0.19	1011	-0.59	1221		
STAAD SPACE											
-- PAGE NO. 56											
MAX/MIN FORCE VALUES FOR MEMB											
FY/	FZ/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD		
MAX.	32.50	0.00	1011	35.54	0.00	1001	54.31	C	0.00	1021	
	0.29	0.00	1214	-2.09	0.00	1011					
MIN.	-30.75	8.02	1013	-34.89	4.81	1011	46.28	T	8.02	1321	
	-0.19	8.02	1011	-1.23	0.00	1214					
5003											
FY/	FZ/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD		
MAX.	0.00	MAX	15.24	1021	26.78	1021	2.98	1121	4.50	1321	
	MIN	-7.41	1321	-8.68	1214	-3.94	1221	-5.86	1021		
0.60	MAX	12.26	1021	18.58	1013	2.98	1221	2.26	1321		
	MIN	-9.30	1321	-8.12	1214	-3.94	1121	-4.20	1021		
1.20	MAX	9.86	1021	19.37	1013	2.98	1221	0.03	1321		
	MIN	-11.79	1121	-7.70	1214	-3.94	1121	-2.54	1021		
1.80	MAX	7.97	1221	23.32	1011	2.98	1221	0.16	1214		
	MIN	-14.77	1121	-7.43	1214	-3.94	1121	-2.76	1121		
2.40	MAX	6.07	1221	30.24	1011	2.98	1221	1.46	1221		
	MIN	-17.75	1121	-7.30	1214	-3.94	1121	-5.12	1121		
3.00	MAX	4.17	1221	39.99	1011	2.98	1221	3.25	1221		
	MIN	-20.73	1121	-7.32	1214	-3.94	1121	-7.48	1121		
5004											
FY/	FZ/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD		
MAX.	15.24	0.00	1021	39.99	3.00	1011	97.66	C	0.00	1221	
	2.98	0.00	1221	4.50	0.00	1321					
MIN.	-20.73	3.00	1121	-8.68	0.00	1214	98.90	T	3.00	1121	
	-3.94	3.00	1121	-7.48	3.00	1121					
5004											
FY/	FZ/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD		
MAX.	0.00	MAX	33.14	1011	45.34	1121	0.39	1011	1.58	1221	
	MIN	-4.68	1214	-8.20	1214	-0.11	1221	-1.50	1121		
1.60	MAX	20.53	1011	15.85	1321	0.39	1011	1.40	1021		
	MIN	-4.03	1214	-18.66	1021	-0.11	1221	-1.04	1321		
3.21	MAX	7.92	1011	4.72	1214	0.39	1011	1.30	1021		
	MIN	-3.28	1214	-26.96	1021	-0.11	1221	-0.65	1321		

Vehicle Maintenance Shop-r1.AWL

5005, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	DIST LD	FX	DIST LD	
MAX.	22.01	0.00	0.00	33.16	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6001									
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6002									
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6003									
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

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6001, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	DIST LD	FX	DIST LD	
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6002									
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6003									
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

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6002, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	DIST LD	FX	DIST LD	
MAX.	2.83	0.00	0.00	7.79	8.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6003									
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6004									
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

§ STAAD SPACE 60 -- PAGE NO. 60

Vehicle Maintenance Shop-r1.AWL

6004, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	DIST LD	FX	DIST LD	
MAX.	4.01	0.00	0.00	10.36	3.00	0.00	0.00	0.00	0.00
MIN.	12.69	3.00	0.00	22.05	0.00	0.00	0.00	0.00	0.00
6005									
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

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6005, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	DIST LD	FX	DIST LD	
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6006									
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

§ STAAD SPACE 62 -- PAGE NO. 62

6006, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	DIST LD	FX	DIST LD	
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6007									
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

§ STAAD SPACE 63 -- PAGE NO. 63

6007, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	DIST LD	FX	DIST LD	
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

§ STAAD SPACE 64 -- PAGE NO. 64

Vehicle Maintenance Shop-r1.ANL									
5.66	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7.55	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
9.44	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
STAAD SPACE									

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--- PAGE NO. 64

MAX/MIN FORCE VALUES FOR MEMB 8003, AMONGST ALL SECT LOCATIONS									
FY/		DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

8004	0.00	MAX	0.53	0.00	0.00	0.00	0.00	0.00	0.00
	1.87	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	3.73	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	5.60	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	7.46	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	9.33	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00

MAX/MIN FORCE VALUES FOR MEMB 8004, AMONGST ALL SECT LOCATIONS									
FY/		DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	-0.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

8005	0.00	MAX	0.53	0.00	0.00	0.00	0.00	0.00	0.00
	1.87	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	3.73	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	5.60	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	7.46	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	9.33	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00

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MAX/MIN FORCE VALUES FOR MEMB 8005, AMONGST ALL SECT LOCATIONS									
FY/		DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	-0.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

8006	0.00	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	1.89	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	3.77	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	5.66	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	7.55	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00

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9.44	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Vehicle Maintenance Shop-r1.ANL

MAX/MIN FORCE VALUES FOR MEMB 8006, AMONGST ALL SECT LOCATIONS									
FY/		DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

8007	0.00	MAX	0.54	0.00	0.00	0.00	0.00	0.00	0.00
	1.89	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	3.78	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	5.67	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	7.55	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	9.44	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00

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STAAD SPACE

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MAX/MIN FORCE VALUES FOR MEMB 8007, AMONGST ALL SECT LOCATIONS									
FY/		DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.54	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	-0.54	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

8008	0.00	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	1.90	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	3.79	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	5.69	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	7.58	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	9.48	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00

MAX/MIN FORCE VALUES FOR MEMB 8008, AMONGST ALL SECT LOCATIONS									
FY/		DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MIN.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

8009	0.00	MAX	0.68	0.00	0.00	0.00	0.00	0.00	0.00
	2.40	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	4.80	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	7.20	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	9.60	MAX	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	12.00	MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00

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STAAD SPACE

--- PAGE NO. 67

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[illegible]

	JOINT DISPLACEMENT (CM)	RADIANS	STRUCTURE TYPE = SPACE
1	0.0000	0.0000	
2	0.0000	0.0000	
3	0.0000	0.0000	
4	0.0000	0.0000	
5	0.0000	0.0000	
6	0.0000	0.0000	
7	0.0000	0.0000	
8	0.0000	0.0000	
9	0.0000	0.0000	
10	0.0000	0.0000	
11	0.0000	0.0000	
12	0.0000	0.0000	
13	0.0000	0.0000	
14	0.0000	0.0000	
15	0.0000	0.0000	
16	0.0000	0.0000	
17	0.0000	0.0000	
18	0.0000	0.0000	
19	0.0000	0.0000	
20	0.0000	0.0000	
21	0.0000	0.0000	
22	0.0000	0.0000	
23	0.0000	0.0000	
24	0.0000	0.0000	
25	0.0000	0.0000	
26	0.0000	0.0000	
27	0.0000	0.0000	
28	0.0000	0.0000	
29	0.0000	0.0000	
30	0.0000	0.0000	
31	0.0000	0.0000	
32	0.0000	0.0000	
33	0.0000	0.0000	
34	0.0000	0.0000	
35	0.0000	0.0000	
36	0.0000	0.0000	
37	0.0000	0.0000	
38	0.0000	0.0000	
39	0.0000	0.0000	
40	0.0000	0.0000	
41	0.0000	0.0000	
42	0.0000	0.0000	
43	0.0000	0.0000	
44	0.0000	0.0000	
45	0.0000	0.0000	
46	0.0000	0.0000	
47	0.0000	0.0000	
48	0.0000	0.0000	
49	0.0000	0.0000	
50	0.0000	0.0000	
51	0.0000	0.0000	
52	0.0000	0.0000	
53	0.0000	0.0000	
54	0.0000	0.0000	
55	0.0000	0.0000	
56	0.0000	0.0000	
57	0.0000	0.0000	
58	0.0000	0.0000	
59	0.0000	0.0000	
60	0.0000	0.0000	
61	0.0000	0.0000	
62	0.0000	0.0000	
63	0.0000	0.0000	
64	0.0000	0.0000	
65	0.0000	0.0000	
66	0.0000	0.0000	
67	0.0000	0.0000	
68	0.0000	0.0000	
69	0.0000	0.0000	
70	0.0000	0.0000	
71	0.0000	0.0000	
72	0.0000	0.0000	
73	0.0000	0.0000	
74	0.0000	0.0000	
75	0.0000	0.0000	
76	0.0000	0.0000	
77	0.0000	0.0000	
78	0.0000	0.0000	
79	0.0000	0.0000	
80	0.0000	0.0000	
81	0.0000	0.0000	
82	0.0000	0.0000	
83	0.0000	0.0000	
84	0.0000	0.0000	
85	0.0000	0.0000	
86	0.0000	0.0000	
87	0.0000	0.0000	
88	0.0000	0.0000	
89	0.0000	0.0000	
90	0.0000	0.0000	
91	0.0000	0.0000	
92	0.0000	0.0000	
93	0.0000	0.0000	
94	0.0000	0.0000	
95	0.0000	0.0000	
96	0.0000	0.0000	
97	0.0000	0.0000	
98	0.0000	0.0000	
99	0.0000	0.0000	
100	0.0000	0.0000	
101	0.0000	0.0000	
102	0.0000	0.0000	
103	0.0000	0.0000	
104	0.0000	0.0000	
105	0.0000	0.0000	
106	0.0000	0.0000	
107	0.0000	0.0000	

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
615	2001	-0.3865	-0.2625	-0.0598	-0.0005	0.0007	-0.0013
	2002	-0.3120	-0.2436	-0.0425	-0.0004	0.0006	-0.0013
	2011	-0.1513	-0.4473	0.3611	0.0002	0.0002	-0.0017
	2012	-0.2822	-0.3115	-0.5530	-0.0002	0.0008	-0.0013
	2013	-0.9604	-0.619	-0.1584	-0.0005	0.0001	-0.0013
2006	2014	-0.4343	-0.2526	-0.0621	-0.0003	0.0005	-0.0013
	2001	0.0000	0.0000	0.0000	0.0001	0.0003	0.0026
	2002	0.0000	0.0000	0.0000	0.0001	0.0003	0.0021
	2011	0.0000	0.0000	0.0000	0.0001	0.0003	0.0021
	2012	0.0000	0.0000	0.0000	0.0013	0.0000	-0.0011
2008	2013	0.0000	0.0000	0.0000	0.0001	0.0002	-0.0011
	2014	0.0000	0.0000	0.0000	0.0001	0.0001	0.0114
	2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002
	2002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002
	2011	0.0000	0.0000	0.0000	0.0015	-0.0001	-0.0005
2010	2012	0.0000	0.0000	0.0000	0.0014	-0.0002	-0.0012
	2013	0.0000	0.0000	0.0000	0.0000	0.0000	0.0083
	2014	0.0000	0.0000	0.0000	0.0000	0.0000	0.0111
	2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0007
	2002	0.0000	0.0000	0.0000	0.0000	-0.0001	0.0009
2011	2011	0.0000	0.0000	0.0000	0.0000	-0.0001	0.0009
	2012	0.0000	0.0000	0.0000	0.0013	-0.0003	0.0003
	2013	0.0000	0.0000	0.0000	0.0013	-0.0003	0.0003
	2014	0.0000	0.0000	0.0000	0.0000	-0.0003	0.0018
	2001	0.0000	0.0000	0.0000	-0.0001	-0.0002	0.0007
2011	2002	0.0000	0.0000	0.0000	-0.0001	-0.0003	0.0026
	2011	0.0000	0.0000	0.0000	0.0021	-0.0001	0.0021
	2012	0.0000	0.0000	0.0000	0.0014	-0.0004	0.0040
	2013	0.0000	0.0000	0.0000	0.0013	-0.0005	0.0013
	2014	0.0000	0.0000	0.0000	-0.0001	-0.0002	0.0115
3008	2001	0.0000	0.0000	0.0000	-0.0001	-0.0001	0.0114
	2002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0041
	2011	0.0000	0.0000	0.0000	0.0000	0.0000	0.0031
	2012	0.0000	0.0000	0.0000	0.0000	0.0000	0.0031
	2013	0.0000	0.0000	0.0000	-0.0004	-0.0002	0.0019
2013	2014	0.0000	0.0000	0.0000	0.0000	0.0000	0.0064
	2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0013
	2002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0053
	2011	0.0000	0.0000	0.0000	0.0002	0.0003	-0.0043
	2012	0.0000	0.0000	0.0000	0.0015	-0.0001	-0.0026
5007	2013	0.0000	0.0000	0.0000	0.0016	-0.0002	0.0011
	2014	0.0000	0.0000	0.0000	0.0002	0.0002	-0.0109
	2001	0.0000	0.0000	0.0000	0.0003	0.0013	-0.0044
	2002	0.0000	0.0000	0.0000	-0.0003	0.0003	0.0003
	2011	0.0000	0.0000	0.0000	0.0013	0.0001	-0.0003
2012	2012	0.0000	0.0000	0.0000	0.0013	0.0001	-0.0003
	2013	0.0000	0.0000	0.0000	0.0013	0.0001	-0.0003
	2014	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0090
	2001	0.0000	0.0000	0.0000	-0.0002	-0.0001	-0.0008
	2014	0.0000	0.0000	0.0000	-0.0002	-0.0012	-0.0199
STAND SPACE							-- PAGE NO.

JOINT DISPLACEMENT (CM)		RADIAN(S)		STRUCTURE TYPE = SPACE			
JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
5009	2001	0.0000	0.0000	0.0000	0.0003	0.0001	0.0004
	2002	0.0000	0.0000	0.0000	0.0002	0.0000	0.0003
	2003	0.0000	0.0000	0.0000	0.0018	0.0000	-0.0002
	2011	0.0000	0.0000	0.0000	0.0017	-0.0001	0.0002
	2012	0.0000	0.0000	0.0000	0.0003	0.0008	-0.0090
5011	2001	0.0000	0.0000	0.0000	0.0002	0.0002	-0.0199
	2002	0.0000	0.0000	0.0000	-0.0002	-0.0003	-0.0034
	2003	0.0000	0.0000	0.0000	-0.0001	-0.0002	-0.0043
	2011	0.0000	0.0000	0.0000	0.0014	-0.0006	-0.0102
	2012	0.0000	0.0000	0.0000			

[illegible][illegible]

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
515	2001	-0.3804	-0.0310	0.1431	-0.0005	0.0007	0.0008
	2002	-0.3055	-0.0280	0.1243	-0.0004	0.0006	0.0008
	2001	-0.1419	-0.0737	0.7250	-0.0002	0.0002	0.0003
	2012	-0.2765	-0.0752	0.8053	-0.0002	0.0008	0.0008
	2013	-0.3556	-0.0931	0.1585	-0.0005	0.0011	0.0008
516	2001	-0.0210	-0.0210	0.0900	0.0000	0.0000	0.0008
	2002	0.0000	0.0000	0.0000	0.0008	0.0007	0.0011
	2003	0.0000	0.0000	0.0000	0.0007	0.0006	0.0008
	2011	0.0000	0.0000	0.0000	0.0027	0.0012	0.0005
	2012	0.0000	0.0000	0.0000	0.0034	0.0008	0.0007
602	2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0003
	2011	0.0000	0.0000	0.0000	0.0000	0.0000	0.0003
	2012	0.0000	0.0000	0.0000	0.0000	0.0000	0.0003
	2013	0.0000	0.0000	0.0000	0.0000	0.0000	0.0003
	2014	0.0000	0.0000	0.0000	0.0000	0.0000	0.0003
604	2001	-0.3829	-0.2634	0.0627	0.0005	-0.0007	-0.0013
	2002	-0.3089	-0.1642	0.0449	0.0004	0.0006	-0.0013
	2011	-0.0105	-0.1947	0.7004	0.0003	-0.0003	-0.0012
	2012	-0.0164	-0.1327	0.6160	0.0008	-0.0002	-0.0010
	2013	-0.9560	-0.2650	0.1620	0.0005	-0.0011	-0.0013
606	2001	-0.4364	-0.2552	0.0659	0.0000	-0.0005	-0.0013
	2002	-0.3652	-0.2316	0.5830	0.0010	0.0018	-0.0008
	2011	-0.2883	-0.1988	0.5545	0.0008	0.0017	-0.0007
	2012	-0.0095	-0.1013	0.0531	0.0003	0.0020	-0.0007
	2013	-0.0672	-0.0446	0.0000	0.0006	0.0007	-0.0007
608	2001	-0.4321	-0.1214	0.0000	-0.0000	0.0000	-0.0000
	2002	-0.3780	-0.0324	0.3073	0.0007	0.0003	-0.0006
	2011	-0.1780	0.0237	-0.0905	-0.0003	0.0003	0.0036
	2012	-0.1432	0.2477	-0.0510	-0.0003	0.0002	0.0077
	2013	-0.0661	1.0287	0.6169	0.0001	0.0001	0.0029
608	2001	-0.1806	1.0356	0.7124	0.0001	-0.0002	-0.0040
	2012	-0.5842	5.6851	0.3650	-0.0004	0.0012	0.0185
	2014	-0.4088	-0.2712	-0.3802	-0.0002	0.0013	-0.0014
	2001	-0.1965	2.9811	0.0905	0.0000	0.0000	0.0098
	2002	-0.1648	2.2721	0.0004	0.0000	0.0000	0.0075
611	2001	-0.1292	3.2428	0.5712	0.0001	0.0002	0.0108
	2002	-0.1307	0.7251	0.6362	0.0000	0.0000	0.0094
	2011	-0.9302	1.2986	0.0018	0.0000	0.0000	0.0000
	2012	-0.9302	1.2986	0.0018	0.0000	0.0000	0.0000
	2013	-0.1799	3.0307	0.0936	0.0003	0.0003	0.0037
2013	2001	-0.1442	2.4241	0.0636	0.0003	-0.0002	0.0077
	2002	-0.2614	5.4094	0.7999	0.0005	-0.0006	0.0176
	2011	-0.1230	1.3554	0.1755	0.0000	-0.0002	0.0041
	2012	-0.1230	1.3554	0.1755	0.0000	-0.0002	0.0041
Page 53							

ISECT: TAPERED	21	Vehicle Maintenance Shop-r1.ANL
ISECT: TAPERED	1001	159.75
ISECT: TAPERED	6001	125.17
ST UA90X90X6		39.463
		39.199
		18.273
		224.40

		166.638

***** END OF DATA FROM INTERNAL STORAGE *****

274. FINISH

***** END OF THE STAAD.PRO RUN *****

**** DATE= APR 28,2007 TIME= 15: 1: 9 ****

2 FOUNDATION DESIGN CHECK

Document No.: **DM001-E-MAP-CDP-DR-DCC-371951**

Rev A4

Date: 02 May 07

Page

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site: Main Depot Rashidiya - Building 23
Footings Summary

Date: Apr '07

Job No. 068010

Designed by Rudy

Sheet No.

Soil Bearing Pressure = 150 kN/m²
Soil Density = 20 kN/m³
Concrete Density = 24 kN/m³
Footings Depth below slab soffit = 1 m
Column Stump Area = 0.25 m²

Individual Footings

Column	Column Reaction (ULS)	Soil Load (kN)	Footings Load (kN)	Total SLS Load (kN)	ULS Load (kN)	Footings size			Area of Footings (m ²)	Soil Pressure (kN/m ²)	Factor	Remarks
						L (m)	B (m)	H (m)				
Perimeter column	263	75	57.6	352	449	2.000	2.000	0.600	4	88	1.71	OK
Internal column	312	110.2	82.944	453	582	2.400	2.400	0.600	5.76	79	1.91	OK

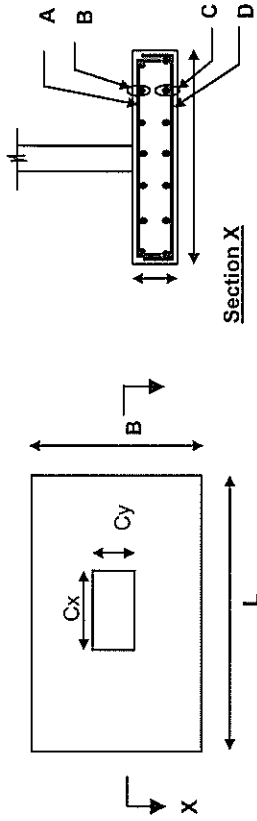
Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 23 Footing Detail	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Summary



Footing Type	L	B	H	Soil Bearing pressure (kPa)	A	Spacing (mm)	B	Spacing (mm)	C	Spacing (mm)	D	Spacing (mm)
F5	2000	2000	600	150	14T12	141	14T12	141	14T16	141	14T16	141
F6	2400	2400	600	150	14T16	172	14T16	172	14T20	172	14T20	172

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 23	Date	Apr '07	Job No.	068010.01
	Footing F5				
	For soil bearing pressure = 150kPa	Designed by	Rudy	Sheet No.	

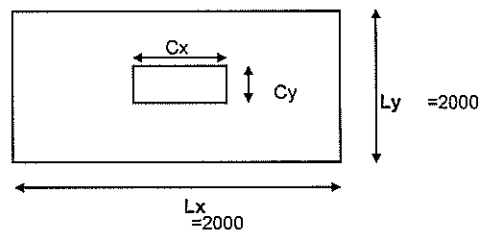
REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	40	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Depth of Footing	h	=	600	mm
Width of Footing	L_y	=	2000	mm
Length of Footing	L_x	=	2000	mm
Width of Column	C_y	=	450	mm
Length of Column	C_x	=	450	mm
Concrete Ten. Cover	c	=	75	mm
Concrete Comp. Cover	c	=	75	mm
Tension bar dia. (x-dir)	D_x	=	16	mm
Tension bar dia. (y-dir)	D_y	=	16	mm
Comp. bar diameter (x-dir)	D_c	=	12	mm
Comp. bar diameter (y-dir)	D_c	=	12	mm
Effect.depth (x-dir)	d_x	=	517	mm
Effect.depth (y-dir)	d_y	=	501	mm
Area of footing	A	=	4	m ²



3. Loadings

Safe bearing pressure	p_b	=	150	KN/m
Load factor	f	=	1.5	
Total axial load (SLS)	N	=	600.00	kN
Eccentricity	e	=	0	mm

4. Analysis

Net pressure (SLS)	p_{max}	=	135.60	KN/m
Net pressure (ULS)	p_{max}	=	203.4	KN/m
Shear force (x-dir)	V_x	=	315.27	kN
Shear force (y-dir)	V_y	=	315.27	kN
Moment (x-dir)	M_x	=	122.17	kNm
Moment (y-dir)	M_y	=	122.17	kNm
Steel required (x-dir)	A_{sreq}	=	1560	mm ²
Steel provided (x-dir)	A_{sprov}	=	2815	mm ²
Steel required (y-dir)	A_{sreq}	=	1560	mm ²
Steel provided (y-dir)	A_{sprov}	=	2815	mm ²
Top bars required (x-dir)	A'_{sreq}	=	1560	mm ²
Top bars provided (x-dir)	A'_{sprov}	=	1583	mm ²
Top bars required (y-dir)	A'_{sreq}	=	1560	mm ²
Top bars provided (y-dir)	A'_{sprov}	=	1583	mm ²

Nos	ϕ	Spacing	
O.K. 14	16	141	x-dir
O.K. 14	16	141	y-dir
O.K. 14	12	141	x-dir
O.K. 14	12	141	y-dir

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 23	Date	Apr '07	Job No.	068010.01
	Footing F5	Designed by	Rudy	Sheet No.	
	For soil bearing pressure = 150kPa				

5. Shear Check

Checking for punching shear:

Shear force at face $V_f = 854.44$ kN
Shear stress at face $v_f = 0.92$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	517	404.49	0.13	0.48	0.96	No Shear Links
2d	1034	0.00	0.00	0.48	0.48	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) $V_c = 348.75$ kN
Shear stress at face (x-dir) $v_c = 0.34$ N/mm² O.K.

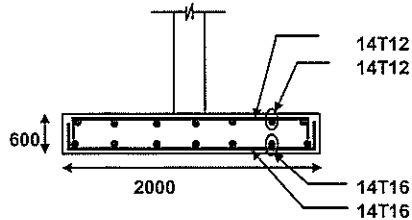
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	517	116.10	0.11	0.48	0.96	No Shear Links
2d	1034	0.00	0.00	0.48	0.48	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir) $V_c = 348.75$ kN
Shear stress at face (y-dir) $v_c = 0.35$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	501	123.30	0.12	0.48	0.97	No Shear Links
2d	1002	0.00	0.00	0.48	0.48	No Shear Links

6. Detailing



Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 23	Date	5/3/2007	Job No.	068010.01
	Footing F6	Designed by	Sebastian	Sheet No.	
	For soil bearing pressure = 150kPa				

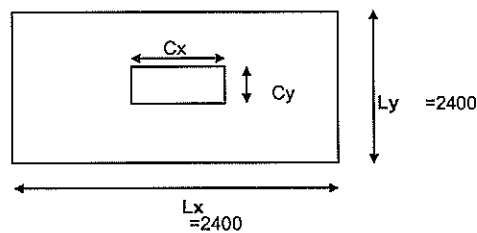
REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	40	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Depth of Footing	h	=	600	mm
Width of Footing	L_y	=	2400	mm
Length of Footing	L_x	=	2400	mm
Width of Column	C_y	=	450	mm
Length of Column	C_x	=	450	mm
Concrete Ten. Cover	c	=	75	mm
Concrete Comp. Cover	c	=	75	mm
Tension bar dia. (x-dir)	D_x	=	20	mm
Tension bar dia. (y-dir)	D_y	=	20	mm
Comp. bar diameter (x-dir)	D_c	=	16	mm
Comp. bar diameter (y-dir)	D_c	=	16	mm
Effect.depth (x-dir)	d_x	=	515	mm
Effect.depth (y-dir)	d_y	=	495	mm
Area of footing	A	=	5.76	m ²



3. Loadings

Safe bearing pressure	p_b	=	150	KN/m
Load factor	f	=	1.5	
Total axial load (SLS)	N	=	864.00	kN
Eccentricity	e	=	0	mm

4. Analysis

Net pressure (SLS)	p_{max}	=	135.60	KN/m
Net pressure (ULS)	p_{max}	=	203.4	KN/m
Shear force (x-dir)	V_x	=	475.96	kN
Shear force (y-dir)	V_y	=	475.96	kN
Moment (x-dir)	M_x	=	232.03	kNm
Moment (y-dir)	M_y	=	232.03	kNm
Steel required (x-dir)	A_{sreq}	=	1872	mm ²
Steel provided (x-dir)	A_{sprov}	=	4398	mm ²
Steel required (y-dir)	A_{sreq}	=	1872	mm ²
Steel provided (y-dir)	A_{sprov}	=	4398	mm ²
Top bars required (x-dir)	A'_{sreq}	=	1872	mm ²
Top bars provided (x-dir)	A'_{sprov}	=	2815	mm ²
Top bars required (y-dir)	A'_{sreq}	=	1872	mm ²
Top bars provided (y-dir)	A'_{sprov}	=	2815	mm ²

	Nos	ϕ	Spacing	
O.K.	14	20	172	x-dir
O.K.	14	20	172	y-dir
O.K.	14	16	172	x-dir
O.K.	14	16	172	y-dir

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 23	Date	5/3/2007	Job No.	068010.01
	Footing F6	Designed by	Sebastian	Sheet No.	
	For soil bearing pressure = 150kPa				

5. Shear Check

Checking for punching shear:

Shear force at face $V_f = 1250.44$ kN
Shear stress at face $v_f = 1.35$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	515	803.16	0.26	0.52	1.05	No Shear Links
2d	1030	0.00	0.00	0.52	0.52	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) $V_c = 526.50$ kN
Shear stress at face (x-dir) $v_c = 0.43$ N/mm² O.K.

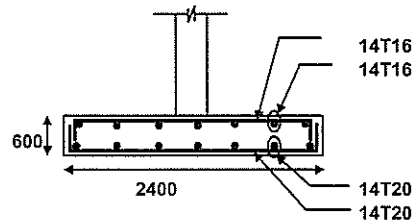
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	515	248.40	0.20	0.52	1.05	No Shear Links
2d	1030	0.00	0.00	0.52	0.52	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir) $V_c = 526.50$ kN
Shear stress at face (y-dir) $v_c = 0.44$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	495	259.20	0.22	0.53	1.06	No Shear Links
2d	990	0.00	0.00	0.53	0.53	No Shear Links

6. Detailing



2.1 GROUND BEAM CHECK

Document No.: **DM001-E-MAP-CDP-DR-DCC-371951**

Rev A4

Date: 02 May 07

Page

PROKON

Software Consultants (Pty) Ltd

Internet: <http://www.prokon.com>

E-Mail: mail@prokon.com

Job Number

068010.21

Job Title

Dubai Depot - Vehicle Maintenance Shop

Client

JTMJV

Collec by

CJA

Checked by

Rudy

Date

Apr '07

Sheet

2

Sections

Beam Elevation

Live

Dead

Over

14.15

164.00

5.000

5.000

5.000

5.000

20.000

$S.D.L - slab = 24 \times 0.3 \times 1 = 7.2 \text{ kN/m}$

$Finishes = 24 \times 0.03 \times 1 = 1.2 \text{ kN/m}$

$LL = 5 \times 1 = 5 \text{ kN/m}$

$P_1 = 180 \text{ kN}$

$P_2 = 164 \text{ kN}$

From STAAD Pro analysis

(ULS Reaction / 1.4)

PROKON

Software Consultants (Pty) Ltd

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Job Number

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Job Title

Dubai Depot - Vehicle Maintenance Shop

Client

JTMJV

Collec by

CJA

Checked by

Rudy

Date

Apr '07

Sheet

1

Continuous Beam: Vehicle Maintenance Building - Ground Beam GL 23-1

5/3/2007 9:14:51 AM

(Code of Practice C01)

Input Tables

Fcu (MPa)	30
Fy (MPa)	460
Fyv (MPa)	460
% Redistribution	0
Downward/Optimized redistr.	0
Cover to centre top steel (mm)	95
Cover to centre bot steel (mm)	95
Dead Load Factor	1.4
Live Load Factor	1.6
Density of concrete (kN/m ³)	24
% Live load permanent	25
Ø (Creep coefficient)	2
Ecs (Free shrinkage strain)	300E-6

Sec No.	Bw (mm)	D (mm)	Bi-top (mm)	Hi-top (mm)	Bi-bot (mm)	Hi-bot (mm)	Y-offset (mm)	Web offset	Flange offset
1	400	600							

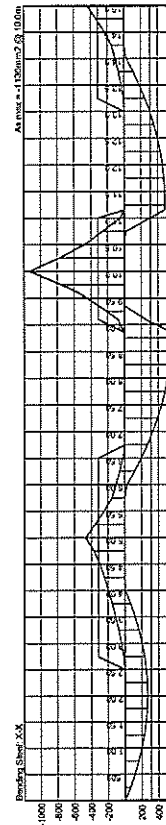
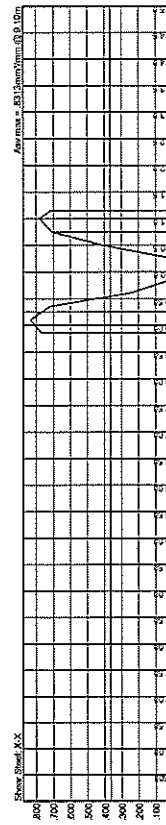
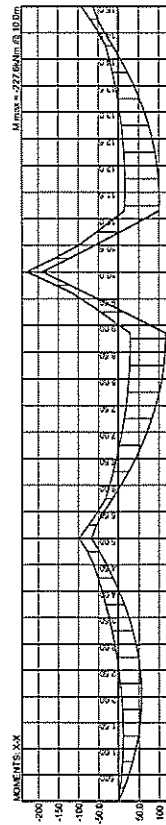
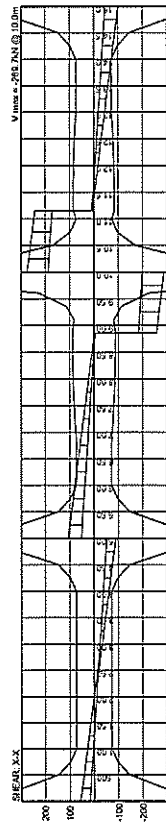
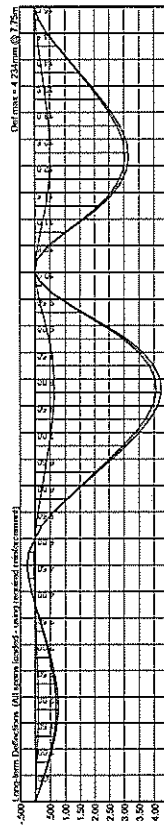
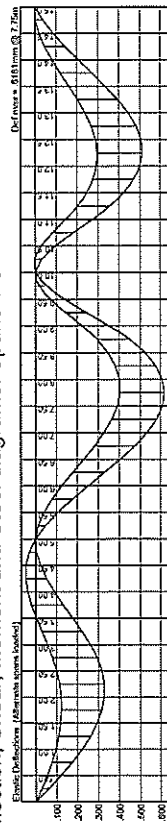
Span No.	Section Length (m)	Sec No Left	Sec No Right
1	5	1	
2	5	1	
3	5	1	
4	5	1	

Sup No.	Code C.F	Column Below D (mm)	B (mm)	H (m)	Code F.P	Column above D (mm)	B (mm)	H (m)	Code F.P

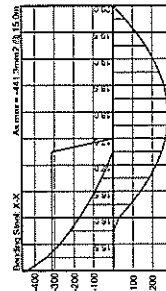
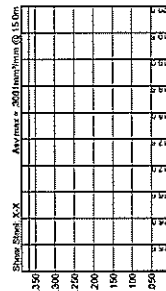
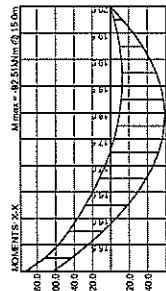
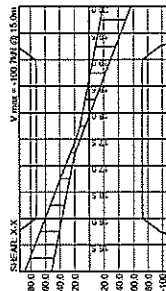
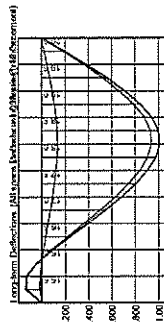
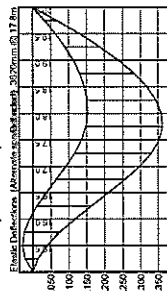
Case D/L	Span	Wleft (kNm)	Wright (kNm)	a (m)	b (m)	P (kN)	a (m)	M (kNm)	a (m)
D	1	8.4	0	20	20				
L	1	5	0						
D	2					180	3.855		
D	3					164	1.149		

Span	M left (kNm)	M right (kNm)

Deflection, Shear, Moment and Steel diagrams. Spans 1-3



Deflection, Shear, Moment and Steel diagrams. Span 4



PROKON		Job Number	068010.21	Sheet	5
Software Consultants (Pty) Ltd		Job Title	Dubai Depot - Vehicle Maintenance Shop		
Internet: http://www.prokon.com		Client	JTMJV		
E-Mail: mail@prokon.com		Cables by	CJA	Checked by	Rudy
				Date	Apr '07
CONTINUOUS BEAM Ver W2.2.23 - 27 Nov 2006 Title : Vehicle Maintenance Building - Ground Beam GL 23-1 Data File : P:\068010-01\civil\Structural\Reshidiya\Building-23_Vehicle Maintenance Shop\Er Date/Time : 5/3/2007 Code of Practice : BS8110 - 1997					
BENDING MOMENTS & REINFORCEMENT					
SPAN 1					
Position (m) Min.Moment (kNm) Max.Moment (kNm) As-Top (mm2) As-Bot (mm2)					
0.000	0.00	0.00	0.00	0.00	0.00
0.250	4.10	13.16	0.00	62.77	0.00
0.500	7.31	24.58	0.00	117.25	0.00
0.750	9.64	34.26	0.00	163.43	0.00
1.000	11.09	42.21	0.00	201.32	0.00
1.250	11.65	48.41	0.00	230.91	0.00
1.500	11.32	52.88	0.00	252.21	0.00
1.750	10.11	55.60	0.00	265.21	0.00
2.000	8.02	56.59	0.00	269.92	0.00
2.250	5.04	55.84	0.00	266.33	0.00
2.500	1.17	53.35	0.00	254.45	0.00
2.750	-3.58	49.12	17.07	234.28	0.00
3.000	-9.21	43.15	43.95	205.80	0.00
3.250	-15.73	35.44	75.05	169.04	0.00
3.500	-23.14	25.99	110.38	123.98	0.00
3.750	-31.43	14.81	149.92	70.62	0.00
4.000	-40.61	1.88	193.68	8.97	0.00
4.250	-50.67	-12.78	241.67	0.00	0.00
4.500	-61.61	-29.19	293.88	0.00	0.00
4.750	-76.95	-47.33	367.05	0.00	0.00
5.000	-98.39	-67.21	469.31	0.00	0.00
SPAN 2					
Position (m) Min.Moment (kNm) Max.Moment (kNm) As-Top (mm2) As-Bot (mm2)					
0.000	-98.39	-67.21	469.31	0.00	0.00
0.250	-73.91	-54.58	352.52	0.00	0.00
0.500	-51.16	-36.50	244.03	0.00	0.00
0.750	-31.97	-14.29	152.51	0.00	0.00
1.000	-22.00	6.19	104.93	29.55	0.00
1.250	-12.91	24.93	61.57	118.93	0.00
1.500	-4.70	41.94	22.43	200.03	0.00
1.750	2.62	57.20	0.00	272.83	0.00
2.000	9.05	70.72	0.00	337.33	0.00
2.250	14.60	82.51	0.00	393.54	0.00
2.500	19.27	92.55	0.00	441.46	0.00
2.750	23.05	100.86	0.00	481.08	0.00
3.000	25.94	107.43	0.00	512.40	0.00
3.250	27.95	112.25	0.00	535.43	0.00
3.500	29.08	115.34	0.00	550.17	0.00
3.750	29.32	116.69	0.00	556.61	0.00
3.855	29.15	116.74	0.00	556.84	0.00
4.105	-16.86	52.62	80.44	251.01	0.00
4.355	-63.77	-13.23	304.15	0.00	0.00
4.605	-115.38	-80.83	550.34	0.00	0.00

PROKON		Job Number	068010.21	Sheet	6
Software Consultants (Pty) Ltd		Job Title	Dubai Depot - Vehicle Maintenance Shop		
Internet: http://www.prokon.com		Client	JTMJV		
E-Mail: mail@prokon.com		Cables by	CJA	Checked by	Rudy
				Date	Apr '07
SPAN 3					
Position (m) Min.Moment (kNm) Max.Moment (kNm) As-Top (mm2) As-Bot (mm2)					
0.000	-227.64	-188.86	1135.62	0.00	0.00
0.250	-160.15	-122.73	774.08	0.00	0.00
0.500	-99.22	-58.34	473.25	0.00	0.00
0.750	-54.56	4.31	260.26	20.58	0.00
1.000	-10.80	65.23	51.49	311.13	0.00
1.149	14.87	100.71	0.00	480.35	0.00
1.399	16.23	101.44	0.00	483.87	0.00
1.649	16.70	100.44	0.00	479.09	0.00
1.899	16.28	97.70	0.00	466.02	0.00
2.149	14.98	93.22	0.00	444.65	0.00
2.399	12.80	87.00	0.00	414.99	0.00
2.500	11.67	84.00	0.00	400.66	0.00
2.750	8.24	75.34	0.00	359.35	0.00
3.000	3.93	64.94	0.00	309.75	0.00
3.250	-1.27	52.80	6.06	251.86	0.00
3.500	-7.35	38.92	35.07	185.67	0.00
3.750	-14.32	23.31	68.30	111.18	0.00
4.000	-22.17	5.95	105.75	28.40	0.00
4.250	-30.91	-13.14	147.42	0.00	0.00
4.500	-48.34	-33.97	230.57	0.00	0.00
4.750	-69.56	-51.03	331.77	0.00	0.00
5.000	-92.51	-62.43	441.26	0.00	0.00
SPAN 4					
Position (m) Min.Moment (kNm) Max.Moment (kNm) As-Top (mm2) As-Bot (mm2)					
0.000	-92.51	-62.43	441.26	0.00	0.00
0.250	-71.36	-42.78	340.39	0.00	0.00
0.500	-56.84	-24.88	271.12	0.00	0.00
0.750	-46.16	-8.72	220.17	0.00	0.00
1.000	-36.36	5.71	173.45	27.22	0.00
1.250	-27.45	18.39	130.95	87.73	0.00
1.500	-19.43	29.34	92.67	139.94	0.00
1.750	-12.29	38.55	58.61	183.86	0.00
2.000	-6.03	46.02	28.78	219.49	0.00
2.250	-0.66	51.75	3.16	246.82	0.00
2.500	3.82	55.74	0.00	265.86	0.00
2.750	7.42	57.99	0.00	276.60	0.00
3.000	10.14	58.50	0.00	279.04	0.00
3.250	11.97	57.28	0.00	273.19	0.00
3.500	12.91	54.31	0.00	259.05	0.00
3.750	12.97	49.61	0.00	236.61	0.00
4.000	12.15	43.16	0.00	205.88	0.00
4.250	10.44	34.98	0.00	166.85	0.00
4.500	7.84	25.06	0.00	119.53	0.00
4.750	4.36	13.40	0.00	63.91	0.00
5.000	0.00	0.00	0.00	0.00	0.00
SHEAR FORCES & REINFORCEMENT					
SPAN 1					



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Internet: <http://www.prokon.com>
E-Mail: mail@prokon.com

Job Number068010.21
Job TitleDubai Depot - Vehicle Maintenance Shop
ClientJTMJV
Dates byCJA
Checked byRudy
DateApr '07

Sheet7

Position (m)	Minimum Shear (kN)	Maximum Shear (kN)	Shear Capacity (kN)	Asv/Sv (mm2/mm)	Asv/Sv (nom) (mm2/mm)
0.000	18.17	56.12	885.12	0.00	0.37
0.250	14.63	49.16	291.21	0.00	0.37
0.500	11.09	42.21	145.61	0.00	0.37
0.750	7.55	35.25	97.07	0.00	0.37
1.000	4.01	28.29	72.80	0.00	0.37
1.250	0.47	21.34	72.08	0.00	0.37
1.500	-3.07	14.38	72.08	0.00	0.37
1.750	-6.61	7.43	72.08	0.00	0.37
2.000	-10.15	0.47	72.08	0.00	0.37
2.250	-13.69	-6.49	72.08	0.00	0.37
2.500	-19.68	-13.44	72.08	0.00	0.37
2.750	-26.63	-20.40	72.08	0.00	0.37
3.000	-33.59	-24.31	72.08	0.00	0.37
3.250	-40.55	-27.85	72.08	0.00	0.37
3.500	-47.50	-31.39	72.08	0.00	0.37
3.750	-54.46	-34.93	72.08	0.00	0.37
4.000	-61.41	-38.47	72.80	0.00	0.37
4.250	-68.37	-42.01	97.07	0.00	0.37
4.500	-75.33	-45.55	145.61	0.00	0.37
4.750	-82.28	-49.09	310.43	0.00	0.37
5.000	-89.24	-52.63	885.12	0.00	0.37

SPAN 2

Position (m)	Minimum Shear (kN)	Maximum Shear (kN)	Shear Capacity (kN)	Asv/Sv (mm2/mm)	Asv/Sv (nom) (mm2/mm)
0.000	52.29	106.26	885.12	0.00	0.37
0.250	48.75	99.31	306.28	0.00	0.37
0.500	45.21	92.35	145.61	0.00	0.37
0.750	41.67	85.40	97.07	0.00	0.37
1.000	38.13	78.44	72.80	0.03	0.37
1.250	34.59	71.48	72.08	0.00	0.37
1.500	31.05	64.53	72.08	0.00	0.37
1.750	27.51	57.57	72.08	0.00	0.37
2.000	23.97	50.62	74.71	0.00	0.37
2.250	20.43	43.66	78.65	0.00	0.37
2.500	16.89	36.70	81.72	0.00	0.37
2.750	13.35	29.75	84.09	0.00	0.37
3.000	9.81	22.79	85.88	0.00	0.37
3.250	6.27	15.84	87.15	0.00	0.37
3.500	2.73	8.88	87.94	0.00	0.37
3.750	-2.92	1.92	88.28	0.00	0.37
3.855	-5.84	-1.00	88.29	0.00	0.37
3.855	-257.84	-182.30	88.29	0.77	0.37
4.105	-264.80	-185.84	81.34	0.83	0.37
4.355	-271.76	-189.38	113.02	0.72	0.37
4.605	-278.71	-192.92	224.88	0.24	0.37
4.855	-285.67	-196.46	724.20	0.00	0.37
5.000	-289.70	-198.51	885.12	0.00	0.37

SPAN 3

Position (m)	Minimum Shear (kN)	Maximum Shear (kN)	Shear Capacity (kN)	Asv/Sv (mm2/mm)	Asv/Sv (nom) (mm2/mm)
0.000	187.46	273.42	885.12	0.00	0.37
0.250	183.92	266.47	398.10	0.00	0.37
0.500	180.38	259.51	168.94	0.41	0.37

PROKON

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E-Mail: mail@prokon.com

Job Number

068010.21

Job Title

Dubai Depot - Vehicle Maintenance Shop

Client

JTMJV

Cuts by

CJA

Checked by

Rudy

Date

Apr '07

Position (m)	Minimum Shear (kN)	Maximum Shear (kN)	Shear Capacity (kN)	Asv/Sv (mm2/mm)	Asv/Sv (nom) (mm2/mm)
0.000	51.57	88.06	885.12	0.00	0.37
0.250	48.03	81.11	302.73	0.00	0.37
0.500	44.49	74.15	145.61	0.00	0.37
0.750	40.95	67.19	97.07	0.00	0.37
1.000	37.41	60.24	72.80	0.00	0.37
1.250	33.87	53.28	72.08	0.00	0.37
1.500	30.33	46.33	72.08	0.00	0.37
1.750	26.79	39.37	72.08	0.00	0.37
2.000	23.25	32.41	72.08	0.00	0.37
2.250	19.44	25.46	72.08	0.00	0.37
2.500	12.49	18.50	72.08	0.00	0.37
2.750	5.53	12.63	72.08	0.00	0.37
3.000	-1.43	9.09	72.08	0.00	0.37
3.250	-8.38	5.55	72.08	0.00	0.37
3.500	-15.34	2.01	72.08	0.00	0.37
3.750	-22.29	-1.53	72.08	0.00	0.37
4.000	-29.25	-5.07	72.80	0.00	0.37
4.250	-36.21	-8.61	97.07	0.00	0.37
4.500	-43.16	-12.15	145.61	0.00	0.37
4.750	-50.12	-15.69	291.21	0.00	0.37
5.000	-57.07	-19.23	885.12	0.00	0.37

SPAN 4

Position (m)	Minimum Shear (kN)	Maximum Shear (kN)	Shear Capacity (kN)	Asv/Sv (mm2/mm)	Asv/Sv (nom) (mm2/mm)
0.750	176.84	252.56	97.07	0.70	0.37
1.000	173.30	245.60	73.45	0.78	0.37
1.149	171.19	241.45	84.05	0.71	0.37
1.149	6.43	11.65	84.05	0.00	0.37
1.399	-0.53	4.90	84.25	0.00	0.37
1.649	-7.48	0.11	83.98	0.00	0.37
1.899	-14.44	-3.43	83.20	0.00	0.37
2.149	-21.40	-6.97	81.91	0.00	0.37
2.399	-28.35	-10.51	80.05	0.00	0.37
2.500	-31.16	-11.94	79.12	0.00	0.37
2.750	-38.12	-15.48	76.30	0.00	0.37
3.000	-45.07	-19.02	72.61	0.00	0.37
3.250	-52.03	-22.56	72.08	0.00	0.37
3.500	-58.99	-26.10	72.08	0.00	0.37
3.750	-65.94	-29.64	72.08	0.00	0.37
4.000	-72.90	-33.18	72.80	0.00	0.37
4.250	-79.85	-36.72	97.07	0.00	0.37
4.500	-86.81	-40.26	145.61	0.00	0.37
4.750	-93.77	-43.80	300.15	0.00	0.37
5.000	-100.72	-47.34	885.12	0.00	0.37

COLUMN REACTIONS

Support	Dead Load (kN)	Live Load (kN)	Dead + Live (kN)	Ultimate (kN)
1	24.41	9.82	34.23	56.12
2	103.53	28.57	132.10	190.66
3	375.70	23.21	398.92	563.13
4	98.32	28.57	126.89	183.36
5	25.25	9.82	35.07	57.07

Sheet 8

PROKON		Job Number	068010.21		Sheet	9
Software Consultants (Pty) Ltd Internet: http://www.prokon.com E-Mail: mail@prokon.com		Job Title	Dubai Depot - Vehicle Maintenance Shop			
Client		Client	JTMJV			
Calcs by		Calcs by	CJA	Checked by	Rudy	Date
					Apr '07	
COLUMN MOMENTS (Ultimate limit state)						
Columns below						
Support No.	Bending Moment (kNm) Min. Max.	Shear Force (kN) Min. Max.				
1	0.00 0.00	0.00 0.00				
2	0.00 0.00	0.00 0.00				
3	0.00 0.00	0.00 0.00				
4	0.00 0.00	0.00 0.00				
5	0.00 0.00	0.00 0.00				
Columns above						
Support No.	Bending Moment (kNm) Min. Max.	Shear Force (kN) Min. Max.				
1	0.00 0.00	0.00 0.00				
2	0.00 0.00	0.00 0.00				
3	0.00 0.00	0.00 0.00				
4	0.00 0.00	0.00 0.00				
5	0.00 0.00	0.00 0.00				

ELASTIC DEFLECTIONS (Alternate spans loaded - positive downward)						
Modulus of elasticity: $E_c = 20 + f_{cu}/5 = 26.0 \text{ Mpa}$						
SPAN 1						
Position (m)	Min. Deflection (mm)	Max. Deflection (mm)				
0.000	-0.00	-0.00				
0.250	0.03	0.06				
0.500	0.05	0.12				
0.750	0.08	0.17				
1.000	0.09	0.22				
1.250	0.11	0.26				
1.500	0.12	0.29				
1.750	0.12	0.32				
2.000	0.12	0.33				
2.250	0.11	0.33				
2.500	0.10	0.32				
2.750	0.10	0.32				
3.000	0.08	0.30				
3.250	0.06	0.26				
3.500	0.03	0.22				
3.750	0.00	0.18				
4.000	-0.02	0.13				
4.250	-0.04	0.09				
4.500	-0.05	0.05				
4.750	-0.04	0.02				
5.000	-0.03	0.00				
	-0.00	-0.00				
SPAN 2						
Position (m)	Min. Deflection (mm)	Max. Deflection (mm)				
0.000	-0.00	-0.00				

PROKON

Software Consultants (Pty) Ltd
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E-Mail: mail@prokon.com

Job Number	068010.21	Steel	10
Job Title	Dubai Depot - Vehicle Maintenance Shop		
Client	JTMJV	Date	Apr '07
Calcs by	CJA	Checked by	Rudy

SPAN 3

Position (m)	Min. Deflection (mm)	Max. Deflection (mm)
0.000	-0.00	0.00
0.250	-0.01	0.03
0.500	0.03	0.10
0.750	0.08	0.18
1.000	0.15	0.27
1.149	0.18	0.33
1.149	0.18	0.33
1.399	0.23	0.40
1.649	0.27	0.45
1.899	0.29	0.49
2.149	0.30	0.51
2.399	0.29	0.51
2.500	0.29	0.51
2.500	0.29	0.51
2.750	0.27	0.48
3.000	0.24	0.45
3.250	0.20	0.40
3.500	0.16	0.34
3.750	0.12	0.27
4.000	0.08	0.21
4.250	0.05	0.14
4.500	0.02	0.09
4.750	0.00	0.04
5.000	-0.00	0.00

SPAN 4

Position (m)	Min. Deflection (mm)	Max. Deflection (mm)
0.000	-0.00	0.00
0.250	-0.02	0.01
0.500	-0.03	0.04
0.750	-0.02	0.07
1.000	-0.01	0.12



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Job Number	068010.21	Steel	11
Job Title	Dubai Depot - Vehicle Maintenance Shop		
Client	JTMJV		
Cables by	CJA	Checked by	Rudy
		Date	Apr '07
1.250	0.01		0.16
1.500	0.04		0.21
1.750	0.06		0.26
2.000	0.09		0.30
2.250	0.12		0.33
2.500	0.14		0.35
2.500	0.14		0.35
2.750	0.15		0.36
3.000	0.15		0.36
3.250	0.15		0.34
3.500	0.14		0.32
3.750	0.13		0.28
4.000	0.11		0.24
4.250	0.09		0.19
4.500	0.06		0.13
4.750	0.03		0.07
5.000	0.00		0.00

Total weight of beam = 115.20 kN			

PROKON

Job Number 068010.21

Job Title Dubai Depot - Vehicle Maintenance Shop

Client JTMJV

Internet: <http://www.prokon.com>

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Date Apr '07

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Job Title Dubai Depot - Vehicle Maintenance Shop

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Date Apr '07

Sections

Beam Elevation

$SPL - SLAB = 24 \times 0.3 \times 1 = 7.2 \text{ kN/m}$
 $FINISHES = 24 \times 0.05 \times 1 = 1.2 \text{ kN/m}$
 $UL = 5 \times 1 = 5 \text{ kN/m}$
 $P_1 = 180 \text{ kN}$ } FROM STAAD PRO ANALYSIS
 $P_2 = 164 \text{ kN}$ } CULS REACTION (1st)

PROKON

Job Number 068010.21

Job Title Dubai Depot - Vehicle Maintenance Shop

Client JTMJV

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Date Apr '07

Job Number 068010.21

Job Title Dubai Depot - Vehicle Maintenance Shop

Client JTMJV

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Date Apr '07

Continuous Beam: Vehicle Maintenance Building - Ground Beam GL 23-4

5/3/2007 9:13:25 AM

Input Tables

Fcu (MPa)	30
Fyk (MPa)	460
Fyk (MPa)	460
% Redistribution	0
Downward/Optimized redistrib.	D
Cover to centre top steel (mm)	95
Cover to centre bot steel (mm)	95
Dead Load Factor	1.4
Live Load Factor	1.6
Density of concrete (kNm ³)	24
% Live load permanent	25
Ø (Creep coefficient)	2
Ecs (Free shrinkage strain)	300E-6

Sec No.	Bw (mm)	D (mm)	Ek-top (mm)	Ek-bot (mm)	Hk-top (mm)	Hk-bot (mm)	Yk-top (mm)	Yk-bot (mm)	Flange offset
1	400	600							

Span No	Section Length (m)	Sec No Left	Sec No Right
1	5	1	
2	5	1	
3	5	1	
4	5	1	

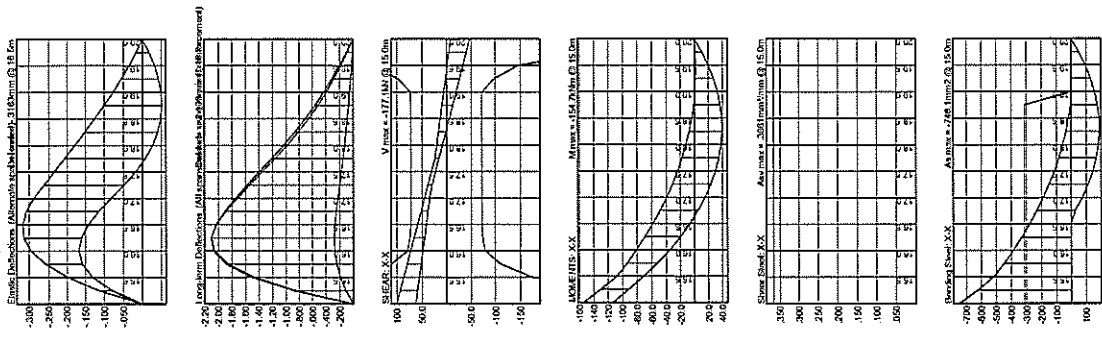
Sup No.	Code C.F	Column Below D (mm)	B (mm)	H (m)	Code F.P	Column above D (mm)	B (mm)	H (m)	Code F.P
---------	----------	---------------------	--------	-------	----------	---------------------	--------	-------	----------

Case D.L	Span	Wleft (kNm)	Wright (kNm)	a (m)	b (m)	P (kN)	a (m)	M (kNm)	a (m)
D	1	8.4		0	20				
L	1	5		0	20				
D	3					180	0		
D	3					164	2.302		

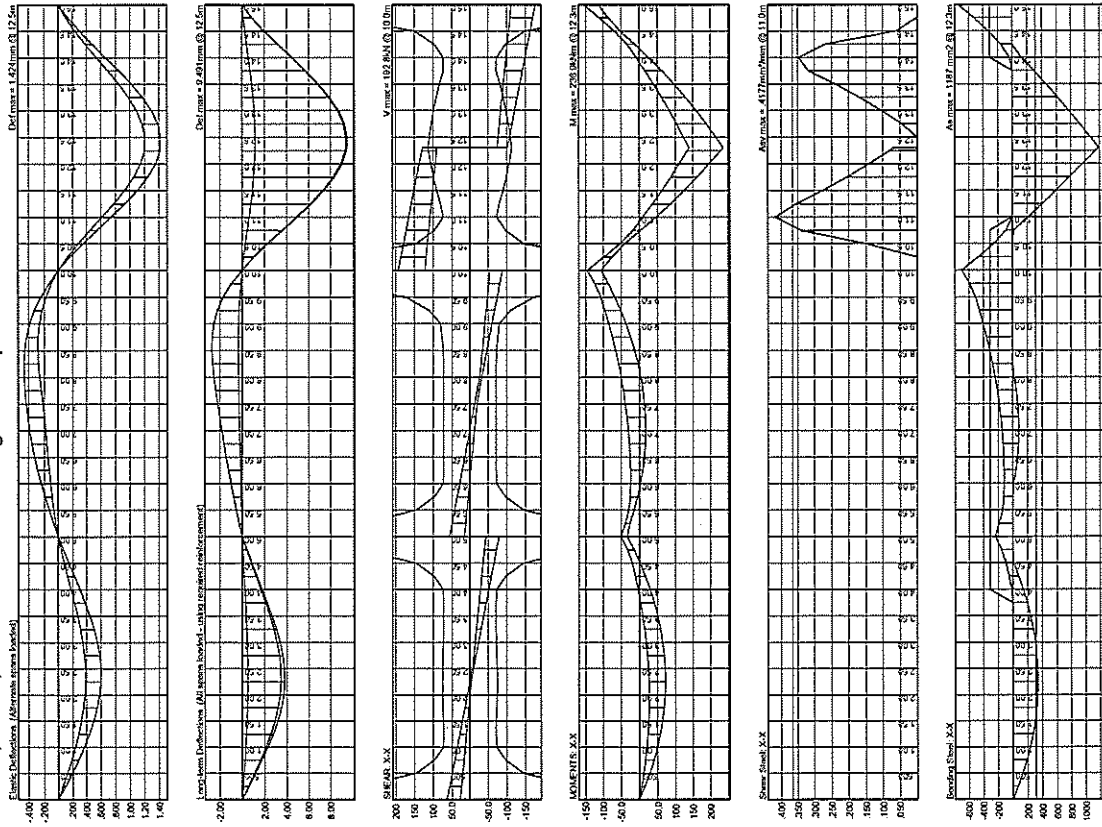
Span	M left (kNm)	M right (kNm)
------	--------------	---------------

062

Deflection, Shear, Moment and Steel diagrams. Span 4



Deflection, Shear, Moment and Steel diagrams. Spans 1-3



PROKON

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Job Number 068010.21

Job Title Dubai Depot - Vehicle Maintenance Shop

Client JTMJV

Calls by CJA

Checked by Rudy

Date Apr '07

Sheet 9

Support No.	Bending Moment (kNm) Min. Max.	Shear Force (kN) Min. Max.
1	0.00 0.00	0.00 0.00
2	0.00 0.00	0.00 0.00
3	0.00 0.00	0.00 0.00
4	0.00 0.00	0.00 0.00
5	0.00 0.00	0.00 0.00

Columns above

Support No.	Bending Moment (kNm) Min. Max.	Shear Force (kN) Min. Max.
1	0.00 0.00	0.00 0.00
2	0.00 0.00	0.00 0.00
3	0.00 0.00	0.00 0.00
4	0.00 0.00	0.00 0.00
5	0.00 0.00	0.00 0.00

ELASTIC DEFLECTIONS (Alternate spans loaded - positive downward)

Modulus of elasticity: $E_c = 20 + f_{cu}/5 = 26.0 \text{ Mpa}$

SPAN 1

Position (m)	Min. Deflection (mm)	Max. Deflection (mm)
0.000	-0.00	-0.00
0.250	0.07	0.10
0.500	0.13	0.20
0.750	0.19	0.29
1.000	0.24	0.37
1.250	0.29	0.44
1.500	0.33	0.50
1.750	0.36	0.55
2.000	0.38	0.58
2.250	0.39	0.60
2.500	0.39	0.60
2.750	0.37	0.59
3.000	0.35	0.55
3.250	0.31	0.51
3.500	0.27	0.45
3.750	0.23	0.38
4.000	0.18	0.31
4.250	0.13	0.23
4.500	0.09	0.15
4.750	0.04	0.07
5.000	0.00	0.00

SPAN 2

Position (m)	Min. Deflection (mm)	Max. Deflection (mm)
0.000	0.00	0.00
0.250	-0.07	-0.03
0.500	-0.13	-0.06
0.750	-0.19	-0.09

PROKON

Software Consultants (Pty) Ltd
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Job Number 068010.21
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Client JTMJV
Calls by CJA
Checked by Rudy
Date Apr '07

Sheet 10

1.000	-0.24	-0.11
1.250	-0.28	-0.13
1.500	-0.32	-0.15
1.750	-0.36	-0.16
2.000	-0.39	-0.18
2.250	-0.41	-0.20
2.500	-0.43	-0.22
2.750	-0.43	-0.22
3.000	-0.45	-0.24
3.250	-0.46	-0.26
3.500	-0.46	-0.27
3.750	-0.45	-0.28
4.000	-0.43	-0.28
4.250	-0.39	-0.26
4.500	-0.33	-0.23
4.750	-0.24	-0.18
5.000	-0.14	-0.10
	-0.00	-0.00

SPAN 3

Position (m)	Min. Deflection (mm)	Max. Deflection (mm)
0.000	-0.00	-0.00
0.250	0.13	0.16
0.500	0.28	0.35
0.750	0.45	0.54
1.000	0.61	0.74
1.250	0.78	0.93
1.500	0.93	1.10
1.750	1.05	1.25
2.000	1.15	1.35
2.250	1.20	1.42
2.302	1.21	1.42
2.500	1.21	1.42
2.500	1.21	1.42
2.750	1.17	1.38
3.000	1.09	1.30
3.250	0.98	1.17
3.500	0.85	1.02
3.750	0.70	0.85
4.000	0.54	0.67
4.250	0.39	0.49
4.500	0.24	0.31
4.750	0.11	0.14
5.000	0.00	0.00

SPAN 4

Position (m)	Min. Deflection (mm)	Max. Deflection (mm)
0.000	0.00	0.00
0.250	-0.11	-0.08
0.500	-0.20	-0.13
0.750	-0.26	-0.16
1.000	-0.29	-0.17
1.250	-0.31	-0.16
1.500	-0.32	-0.14
1.750	-0.31	-0.11
2.000	-0.29	-0.08
2.250	-0.26	-0.05

PROKON Software Consultants (Pty) Ltd Internet: http://www.prokon.com E-Mail: mat@prokon.com		Job Number 068010.21	Sheet 11
Job Title Dubai Depot - Vehicle Maintenance Shop			
Client JTMJV			
Calcs by CJA	Checked by Rudy	Date	Apr '07
2.500	-0.24		
2.500	-0.24		
2.750	-0.21		
3.000	-0.18		
3.250	-0.15		
3.500	-0.13		
3.750	-0.10		
4.000	-0.08		
4.250	-0.05		
4.500	-0.03		
4.750	-0.02		
5.000	-0.00		

Total weight of beam = 115.20 kN			

3 CONNECTIONS DESIGN

Document No.: **DM001-E-MAP-CDP-DR-DCC-371951**

Rev A4

Date: 02 May 07

Page

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site Dubai Depot
Maintenance Vehicle Shop

Date Apr '07 Job No. 068010

Designed by Rudy Sheet No.

Column Splice Connection Design

From the STAAD Pro analysis

Fxc = 107 kN (compression)
Fxt = 0 kN (tension)
Fy = 27 kN
Mz = 105 kNm

Total Tension

$T = (n \cdot T_b / d) \cdot [(d-a) + (d-y_1-a) + (d-y_2-a) + \dots]$
where n = no of row of bolts and T_b is bolt tension capacity

Total Compression

$= 0.5a \cdot p_c \cdot b$
where b = stiff bearing length
= web thickness + 2 * plate thickness

Moment Capacity

$M_c = [n T_b / (d-a)] \cdot [(d-a)^2 + (d-a-y_1)^2 + (d-a-y_2)^2 + \dots]$

For Gr 8.8 bolt

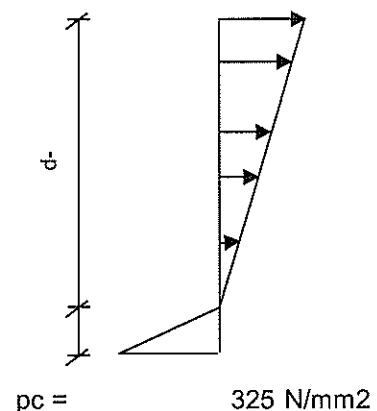
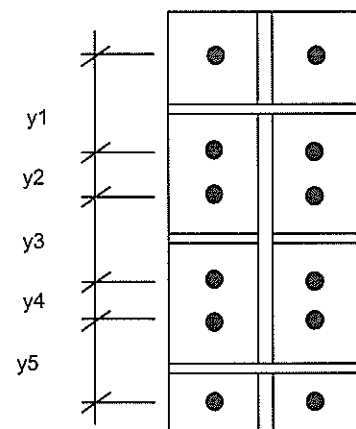
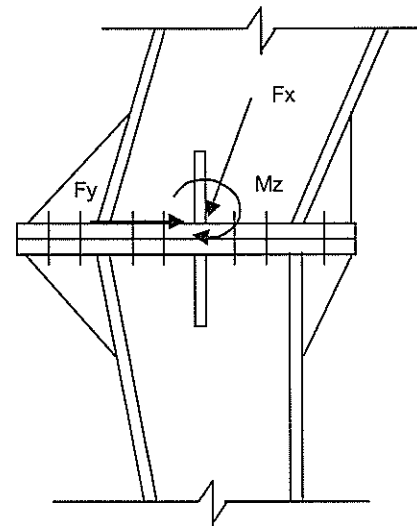
Bolt size = 20 mm
 T_b = 110 kN
 V_b = 91.9 kN
n = 2
n total = 14

web, t_w = 8 mm
plate, t_p = 20 mm
b = 48 mm

y1	125	Trial a	=	78 mm
y2	50	T	=	607.6 kN
y3	100	C	=	607.6 kN
y4	50	T-C	=	0.00 kN
y5	155	M_c	=	170.8 kNm
y6	0			
Depth, d	480			

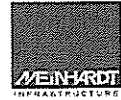
Tension force on the bolt, T_1 = $(M_z / M_c) \cdot T_b$
= 68 kN

$[T_1 + (F_{xt} / n_{total})] / T_b + F_y / (n_{total} \cdot V_b) = 0.64$
< 1.4 Ok



Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai Depot Maintenance Vehicle Shop	Date	Apr '07	Job No.	068010
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Designed by	Rudy	Sheet No.
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Check the effective weld length on the stiffener plate to resist bolt's tension

Weld size, s =	6 mm	(up to 15mm thick)
Weld strength, pw =	1.05 kN/mm	(based on longitudinal capacity)
Weld length required =	$(T1 + 0.5 \cdot F_{xt}) / pw$	
=	64 mm	
=	32 mm per face	

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site Dubai Depot
Maintenance Vehicle Shop

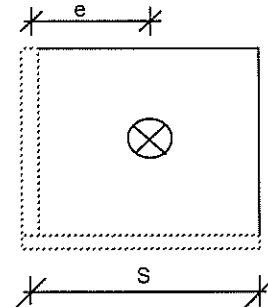
Date Apr '07 Job No. 068010

Designed by Rudy Sheet No.

From the plate analysis (refer attached) using

Prying Force, $T_o = 10$ kN

Plate width, S_o (mm) Ave. support bending moment, M_o (kNm)
150*100 0.114



Actual - Prying force, $T = 68$ kN
Side, $S = 150$ mm
Bending Moment, $M = (T/T_o) * M_o = 0.778$ kNm

Bolt Diameter, $\phi = 20$ mm
Bolt distance, $e = 75$ mm
Thickness of plate, $t_p = 20$ mm
 p_y allow $= 345$ N/mm²

Effective width of the plate resisting the bending moment, $b_p = \min(2e + \phi, S) = 150$ mm

Section of modulus of the plate, $Z_p = b_p * t_p^2 / 6 = 10000$ mm²

$M / Z_p = 78$ N/mm² Ok

Calculation/Sketch

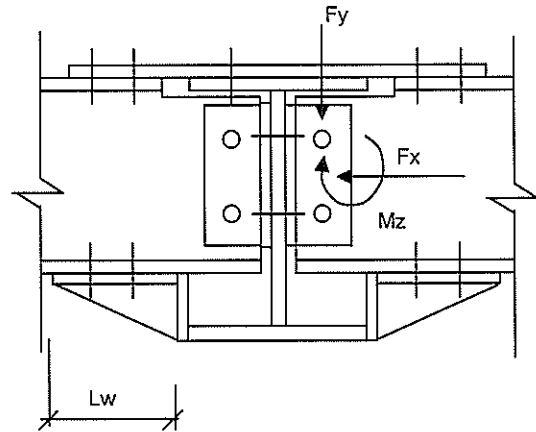
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Site	Dubai Depot Vehicle Maintenance Shop	Date	Apr '07	Job No.	068010
			Designed by	Rudy	Sheet No.

Secondary Beam Connection Design

Roof SDL	=	0.8 kN/m ²
Self weight	=	0.5 kN/m ²
WL	=	0.6 kN/m ² (downward)
WL	=	0.6 kN/m ² (downward)
	=	1.4 kN/m ² (upward)
wuls	-	1.4*DL+1.6LL = 2.78 kN/m ²
	-	1.2*(DL+LL+WL) = 3.00 kN/m ²
	-	1.4WL - DL = 0.66 kN/m ²
loading width, s	=	5 m



For internal span (simply supported span)

- top and bottom plate not required

max span, L = 8 m

Axial load for earthquake, Pa = (11.5+11.5)*(7.1) / 12 rows of secondary beams / tie beams
= 13.6 kN

Vmax = 0.5*wuls*L*s
Downward forces = 60.0 kN
Upward forces = 13.2 kN

Bolt diameter, ϕ = 16 mm
Shear capacity, Vb = 58.9 kN
Bolt tensile capacity, Tb = 70.7 kN

Check shear on bolt connected to secondary beam's web

No of bolt, n = 2
Vmax / n = 30 kN
Pa / n = 6.8 kN
Resultant shear, V = 30.8 kN Ok

Check tensile capacity of bolt connected to main beam's web

Eccentricity of load, e = 60 mm
Moment due to eccentricity, Me = Vmax*e
= 3.6 kNm
Spacing between bolt, db = 150 mm
Tensile force on the bolt, T = Me/db
= 24.0 kN Ok

Calculation/Sketch

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Site	Dubai Depot Vehicle Maintenance Shop	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

For the cantilever end

- Assume shear due to bending is resisted by top and bottom bolt
- Assume vertical and horizontal shear is resisted by bolt on the web

$$\begin{aligned}\text{Span length, } L &= 4 \text{ m} \\ M_{\max} &= 0.5 \cdot w_{\text{uls}} \cdot L^2 \cdot s \\ \text{Downward} &= 120 \text{ kNm} \\ \text{Upward} &= 26.4 \text{ kNm}\end{aligned}$$

$$\begin{aligned}V_{\max} &= w_{\text{uls}} \cdot L \cdot s \\ \text{Downward} &= 60 \text{ kN} \\ \text{Upward} &= 13.2 \text{ kN}\end{aligned}$$

Check bolts on the flanges

$$\begin{aligned}\text{Beam depth - flange thickness, } D_f &= 292 \text{ mm} \\ M_{\max} \text{ bending moment (incl. moment due to ecc), } M_e &= 123.6 \text{ kNm}\end{aligned}$$

$$\begin{aligned}\text{Shear on the flange, } V_f &= M_e / D_f \\ &= 423.3 \text{ kN}\end{aligned}$$

$$\begin{aligned}\text{No of bolt provided} &= 4 \\ \text{Shear per bolt} &= 105.8 \text{ kN} \\ \text{Bolt diameter, } \phi &= 24 \text{ mm} \\ \text{Shear capacity, } V_b &= 132 \text{ kN}\end{aligned}$$

Check weld length required

$$\begin{aligned}\text{Weld size, } s &= 6 \text{ mm} && \text{(up to 15mm thick)} \\ \text{Weld strength, } p_w &= 1.05 \text{ kN/mm} && \text{(based on longitudinal capacity)} \\ \text{Weld length required} &= V_f / p_w \\ &= 403 \text{ mm} \\ &= 202 \text{ mm per face}\end{aligned}$$

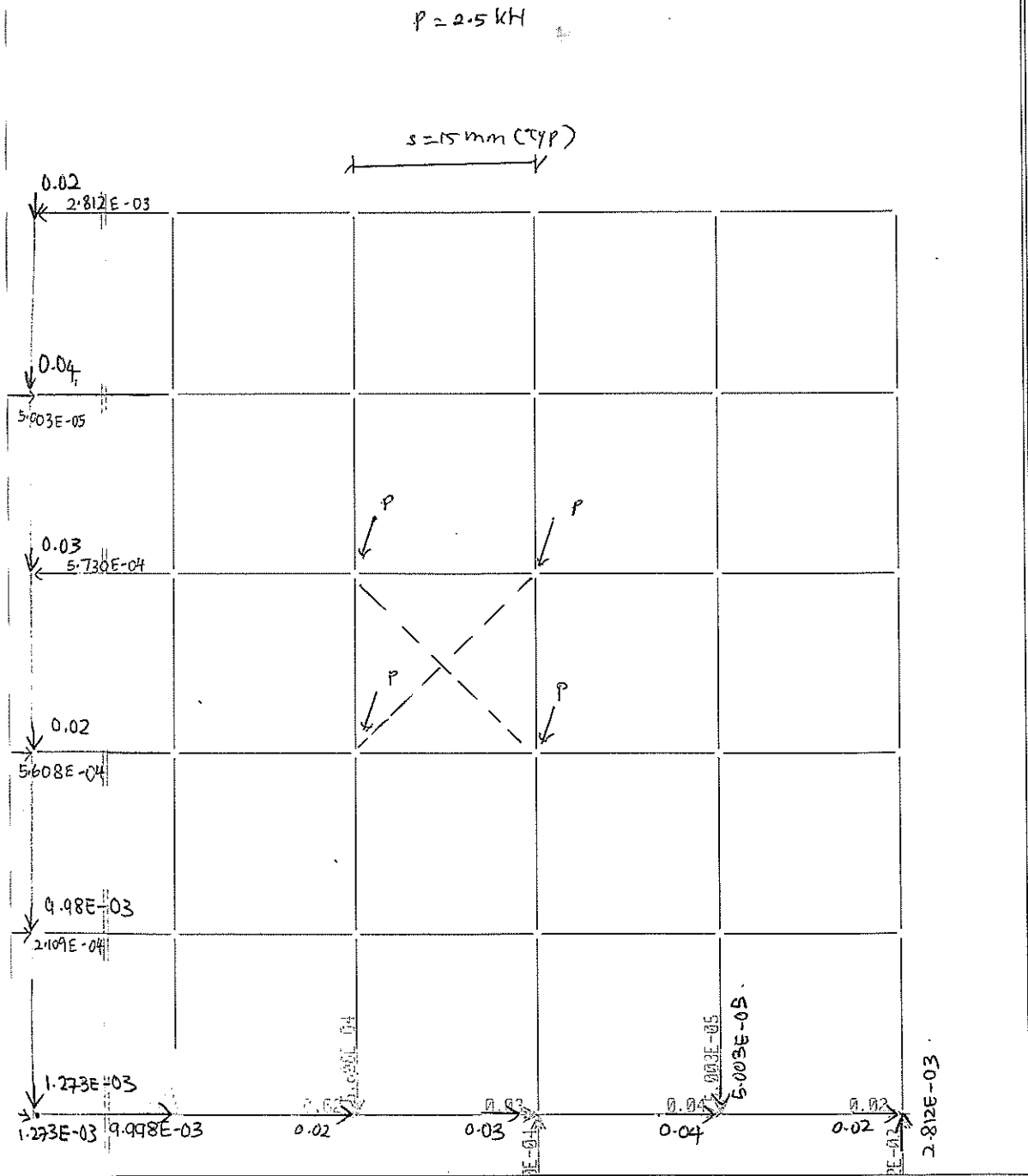
Check shear on the web

- for the secondary beam web

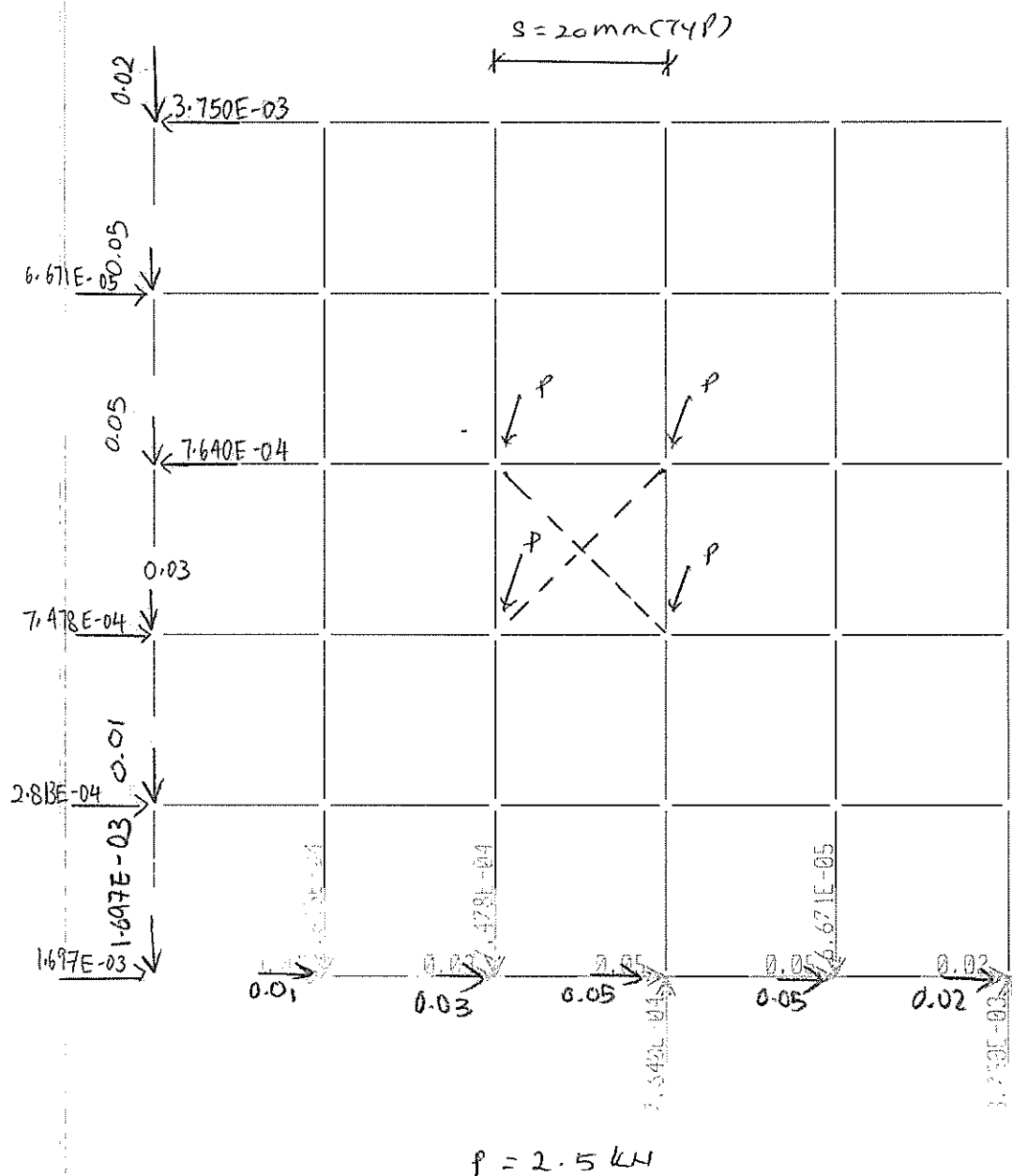
$$\begin{aligned}\text{No of bolt provided} &= 2 \\ \text{Shear per bolt} &= 30.0 \text{ kN} \\ \text{Bolt diameter, } \phi &= 16 \text{ mm} \\ \text{Shear capacity, } V_b &= 58.9 \text{ kN}\end{aligned}$$

- for the main beam web

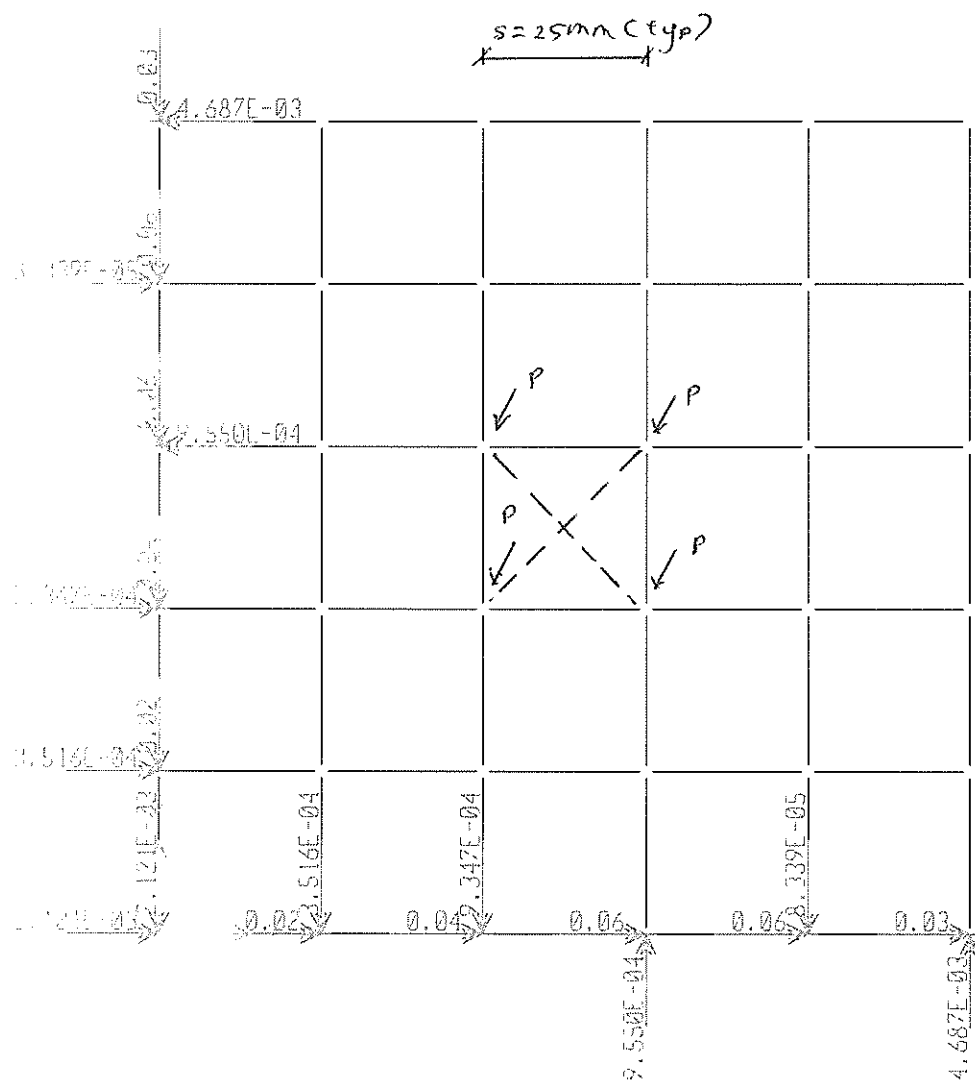
$$\begin{aligned}\text{No of bolt provided} &= 2 \\ \text{Distance between bolt} &= 150 \text{ mm} \\ \text{Tensile load per bolt} &= 24.0 \text{ kN} \\ \text{Bolt diameter, } \phi &= 16 \text{ mm} \\ \text{Tensile capacity, } V_b &= 70.7 \text{ kN}\end{aligned}$$



Bolt Prying Action on Splice Plate
(Plate Width = 75 mm)

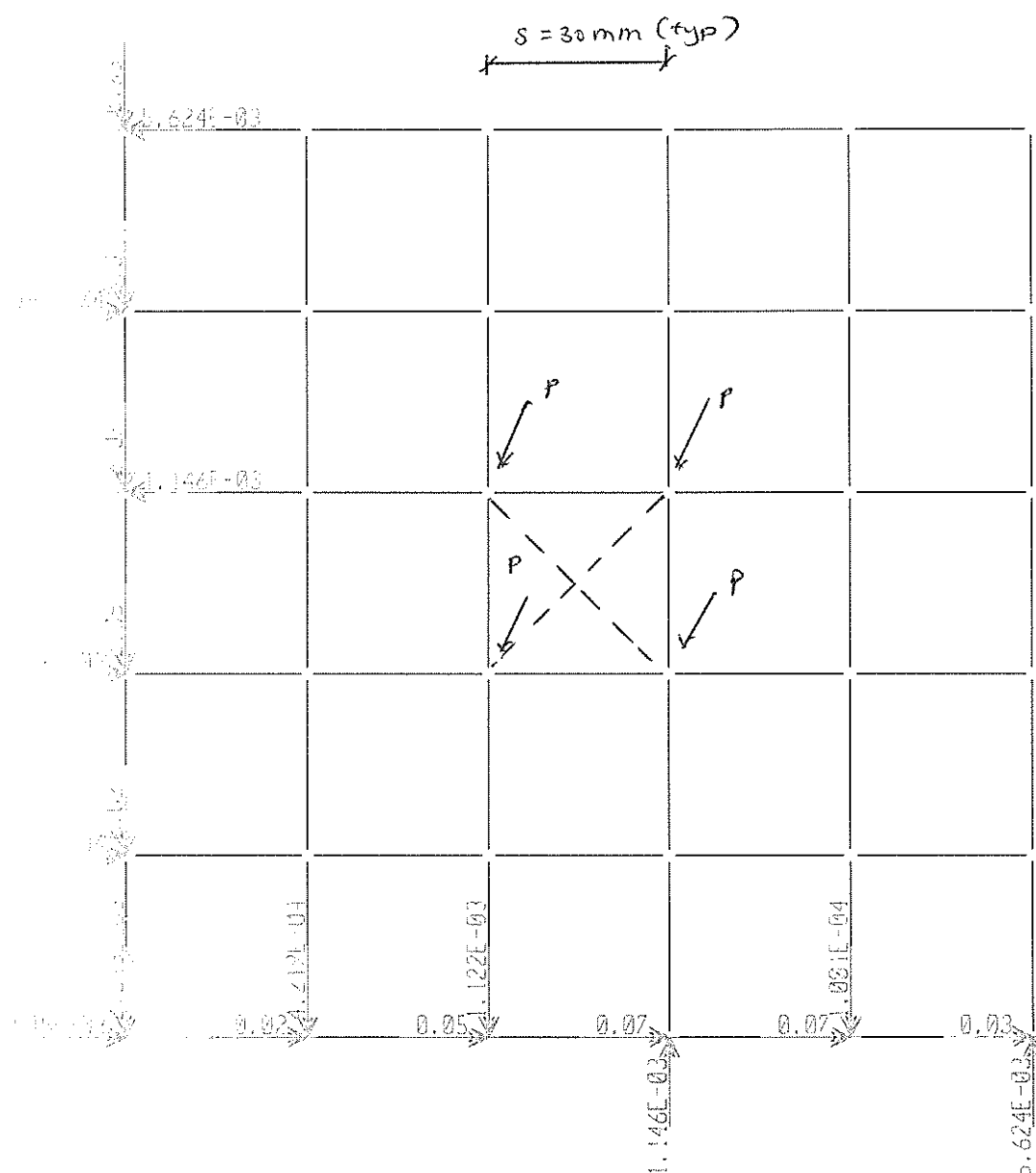


BOLT PRYING ACTION ON SPLICE PLATE
(PLATE WIDTH = 100mm)



$$P = 2.5 \text{ kN}$$

BOLT PRYING ACTION ON SPLICE PLATE
(PLATE WIDTH = 125mm)



$$P = 2.5 \text{ kN}$$

BOLT PRYING ACTION ON SPLICE PLATE
(PLATE WIDTH = 150mm)

